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Hair-bearing human skin generated entirely from pluripotent stem cells

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The skin is a multilayered organ, equipped with appendages (that is, follicles and glands), that is critical for regulating body temperature and the retention of bodily fluids, guarding against external stresses and mediating the sensation of touch and pain[1,2]. Reconstructing appendage-bearing skin in cultures and in bioengineered grafts is a biomedical challenge that has yet to be met[3–9]. Here we report an organoid culture system that generates complex skin from human pluripotent stem cells. We use stepwise modulation of the transforming growth factor β (TGFβ) and fibroblast growth factor (FGF) signalling pathways to co-induce cranial epithelial cells and neural crest cells within a spherical cell aggregate. During an incubation period of 4–5 months, we observe the emergence of a cyst-like skin organoid composed of stratified epidermis, fat-rich dermis and pigmented hair follicles that are equipped with sebaceous glands. A network of sensory neurons and Schwann cells form nerve-like bundles that target Merkel cells in organoid hair follicles, mimicking the neural circuitry associated with human touch. Single-cell RNA sequencing and direct comparison to fetal specimens suggest that the skin organoids are equivalent to the facial skin of human fetuses in the second trimester of development. Moreover, we show that skin organoids form planar hair-bearing skin when grafted onto nude mice. Together, our results demonstrate that nearly complete skin can self-assemble in vitro and be used to reconstitute skin in vivo. We anticipate that our skin organoids will provide a foundation for future studies of human skin development, disease modelling and reconstructive surgery.

Culture systems that contain epidermal and dermal cells have been used to model human skin outside of the body for research or clinical uses for over 40 years[7,8,10]; however, growing functional skin appendages in these systems has been persistently difficult. A previous study reported that appendage-bearing skin organoids can be generated by directed differentiation of epidermal and dermal cells from mouse pluripotent stem cells[3]. From this study, we found that a three-dimensional cyst composed of surface ectodermal cells (that is, epidermal precursor cells) enveloped by mesenchymal cells (that is, dermal precursor cells) can self-organize and mimic mechanisms of cell-to-cell signalling that are critical for the development of hair follicles.

Induction of human skin organoids

Here, we sought to co-induce surface ectodermal and mesenchymal cells from human pluripotent stem cells (Fig. 1a). As a starting point, we revisited a previous method for the generation of inner-ear organoids from human pluripotent stem cells[9], which yields epidermal keratinocytes and mesenchymal tissues as off-target by-products. Human embryonic stem cells of the WA25 cell line were dissociated into single cells and plated on U-bottom 96-well plates to create uniform cell aggregates. We then transferred the aggregates to new plates containing a differentiation medium with key factors to promote epidermal induction: Matrigel, bone morphogenetic protein 4 (BMP4) and a TGFβ inhibitor, SB431542 (Fig. 1b, Extended Data Fig. 1a, Supplementary Note). Inhibition of TGFβ promotes the induction of ectoderm from pluripotent stem cells, whereas BMP4 promotes the induction of surface ectoderm (or non-neural ectoderm) and suppresses the induction of neuroectoderm[9–12]. This differentiation strategy produced uniform epithelial cysts with a diameter of around 500–1,000 μm (Extended Data Fig. 1c). The results could be replicated using a desmoplakin (DSP)-GFP human induced pluripotent stem cell line, in which desmosomes in the epithelium express GFP (Extended Data Fig. 1c). However, in long-term culture (more than 60 days), epidermal cysts never exhibited higher-order

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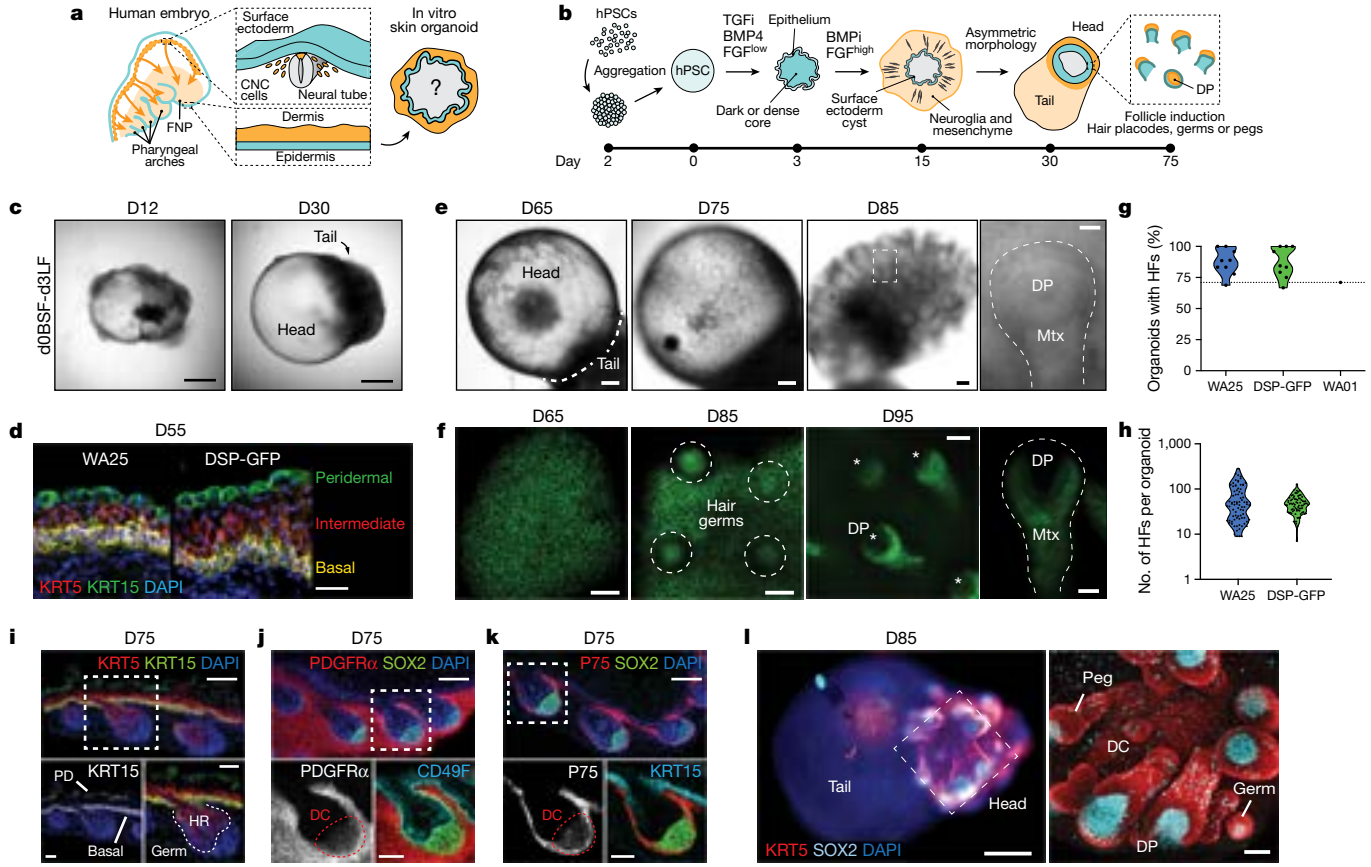


Fig. 1 | Co-induction of surface ectoderm and CNC cells leads to the generation of hair-bearing skin. a, **b**, Overview of study objectives (**a**) and skin organoid protocol (**b**). BMPi, BMP inhibitor; hPSCs, human pluripotent stem cells; DP, dermal papilla; FNP, frontonasal prominence; TGFi, TGFβ inhibitor. **c**, Bright-field images of WA25 aggregates on day (D) 12 and day 30 in optimized culture (that is, treatment with both LDN and FGF; d0BSF-d3LF). Scale bars, 500 µm. **d**, Immunostaining for KRT5$^+$KRT15$^+$ basal and KRT15$^+$ peridermal layers at day 55. Scale bar, 50 µm. **e**, Representative bright-field images of hair follicle induction in WA25 skin organoids at days 65–85. Scale bars, 100 µm (left three panels); 25 µm (far right). **f**, Representative maximum-intensity confocal images (endogenous DSP-GFP) of hair follicle induction in DSP-GFP skin organoids at days 65–95. In **e**, **f**, dashed box, magnified hair follicle; dashed line, hair follicle; dashed circles, developing hair germs; asterisks, dermal papillae. Mtx, matrix. Scale bars, 100 µm (third panel); 50 µm (second panel); 25 µm (far left and far right). **g**, Violin plots showing the frequency of hair follicle (HF) formation in WA25 cultures (n = 212 organoids), DSP-GFP cultures (n = 212 organoids) and WA01 cultures (n = 130 organoids) at days 75–147. **h**, Violin plots showing the average number of hair follicles formed per organoid in WA25 cultures (n = 80 organoids) and DSP-GFP cultures (n = 80 organoids) at days 75–147. **i**–**k**, Immunostained day-75 WA25 skin organoid with hair placodes. Antibodies highlight epidermal (KRT5$^+$KRT15$^+$CD49F$^+$) and peridermal (KRT15$^+$) layers, dermis (PDGFRα$^+$P75$^+$) and dermal condensate cells (SOX2$^+$PDGFRα$^+$P75$^+$). Dashed boxes, magnified regions. Scale bars (**i**–**k**), 100 µm (top); 50 µm (bottom). **l**, Whole-mount sample of day-85 WA25 skin organoid with head and tail structures. KRT5 highlights the epidermis and the outer root sheath of the hair follicle; SOX2 marks dermal condensate cells, dermal papilla cells, Merkel cells and melanocytes. Dashed box, magnified region (right). PD, periderm; HR, hair root; DC, dermal condensate. Scale bars, 250 µm (left); 50 µm (right). See 'Statistics and reproducibility' in Methods for plot and experimental information.

skin morphology, such as stratified keratinocyte layers. Therefore, we looked for stimuli that could be used to co-induce fibroblasts.

Dermal fibroblasts have two embryonic origins: mesodermal cells (body and scalp skin) and cranial neural crest (CNC) cells (facial skin)[13,14]. In our inner-ear culture system, we determined that a co-treatment of basic FGF (FGF) and a BMP inhibitor, LDN-193189 (LDN), promotes the induction of CNC-like cells that can generate mesenchymal cells. However, we did not previously investigate dermal cell specification in the system[9]. Here, we optimized the timing of the LDN and FGF treatment to day 3 of differentiation, and reproducibly generated epithelial cysts that were enveloped by CNC-like cells (Fig. 1c, Extended Data Figs. 1c, 2). To mature the organoids, we incubated them with constant agitation on an orbital shaker. By day 16, the LDN-and-FGF-treated organoids consisted of an inner mantle of TFAP2A$^+$ECAD$^+$ epithelial cells surrounded by an outer layer of migratory TFAP2A$^+$ CNC-like cells (Extended Data Fig. 2). Within the cell mass, there appeared to be two subtypes of CNC-like cells, expressing either the mesenchyme-associated marker PDGFRα or the neuroglia-associated markers SOX10 and P75[15,16] (Extended Data Fig. 2). Clearly visible from day 18, both the WA25 and the DSP-GFP organoids gradually became bipolar, with the epidermal cyst partitioned to one pole (hereafter, the head) and an opaque cell mass at the opposite pole (hereafter, the tail) (Fig. 1c, Extended Data Fig. 1c). After around 50 days of culture, the epithelium stratified into KRT5$^+$KRT15$^+$ basal, KRT5^{low} intermediate (suprabasal) and KRT15$^+$ periderm-like layers—indicative of higher-order skin morphogenesis (Fig. 1d).

Human hair follicles typically develop between weeks 8 and 10 of gestation (Extended Data Fig. 1b); thus, we waited over 70 days for skin organoids to reach a hair-bearing stage. Notably, after a period of quiescence (days 50–70), hair-germ-like buds began extending radially outward from the surface of organoids (Fig. 1e, f, Extended Data Fig. 3). On average, hair germs appeared at 70 ± 5 days for WA25 and 72 ± 4 days for DSP-GFP organoids. Over a period of 120 days, we observed hair follicle induction from batch to batch and across cell lines (87.4% WA25 and 87.2% DSP-GFP organoids with hair follicles) (Fig. 1g, Extended Data Fig. 3a, b, Supplementary Table 1a). The generation of hair-bearing skin organoids was not restricted to these specific cell lines, or to the Koehler laboratory. To validate the protocol, we also generated hair-bearing skin organoids from the WA01 (H1) human embryonic stem cell line

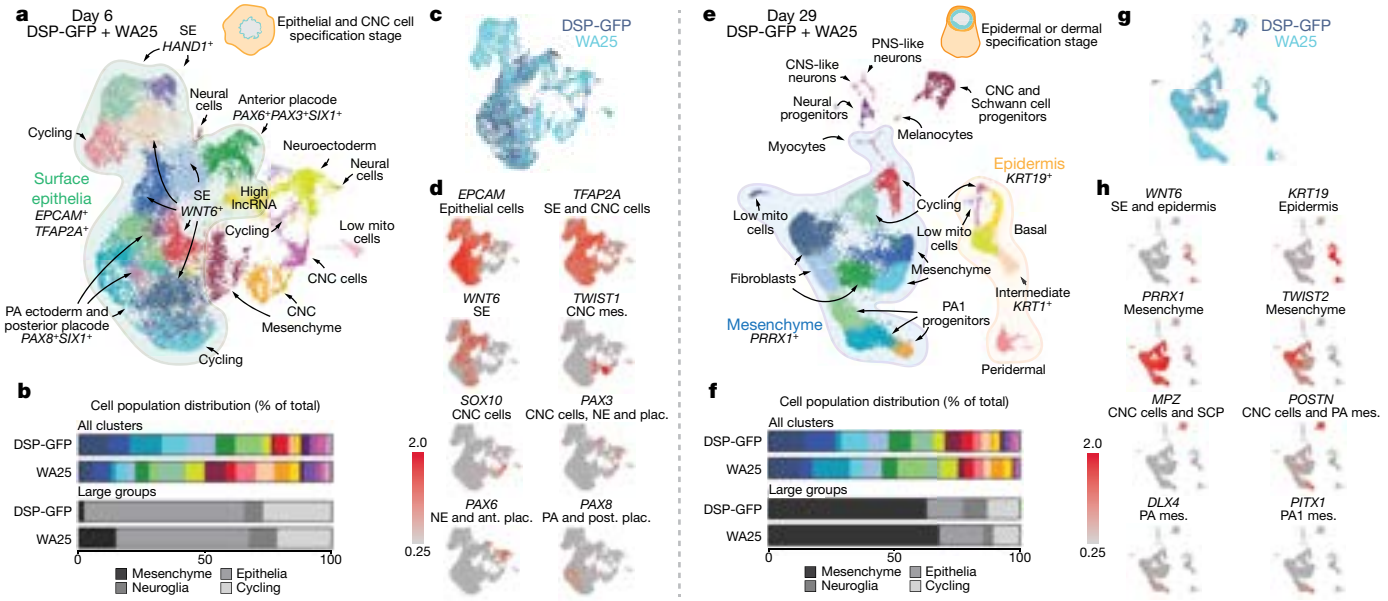


Fig. 2 | scRNA-seq reveals gene expression signatures of craniofacial skin development. a, **e**, Uniform manifold approximation projection (UMAP) plots of day-6 and day-29 WA25 and DSP-GFP integrated datasets. Data represent 23,239 cells (**a**) and 18,190 cells (**e**). The major cell cluster groupings of surface epithelia, epidermis and mesenchyme and the presumptive cell identities (based on a priori knowledge) are noted. Clusters with no discernible identity had either low expression of mitochondrial genes (labelled as 'low mito cells') or high expression of long noncoding RNA genes (labelled as 'high lncRNA'). CNS, central nervous system; PA, pharyngeal arch; PNS, peripheral nervous system; SE, surface ectoderm. **b**, **f**, Percentage distribution of all cell clusters (colour-coded as in **a**, **e**) and of large groups consisting of mesenchymal, epithelial, neuroglial and cycling cells. **c**, **g**, Overlay UMAP plots comparing cells from the DSP-GFP and WA25 datasets. **d**, **h**, Key gene markers for the classification of cell subtype and determination of anterior–posterior patterning. The colour bar to the left shows the average log-normalized expression scale. Ant. plac., anterior placode; mes., mesenchyme; NE, neural ectoderm; PA1, first pharyngeal arch; plac., placode; post. plac., posterior placode.

in the Heller laboratory (70.8% organoids with hair follicles) (Fig. 1g, Extended Data Fig. 3c). By manual counting, we estimated that there were on average 64 hair follicles per WA25 organoid and 48 hair follicles per DSP-GFP organoid (Fig. 1h). As in nascent mouse hair follicles, immunostaining revealed that epidermal germ cells expressed LHX2, PCAD and EDAR and dermal condensates expressed SOX2, PDGFRα and P75 (Fig. 1i–l, Extended Data Fig. 4). The transcription factor SOX2 was expressed in every dermal papilla cell cluster examined (Fig. 1i–l, Supplementary Videos 1, 2), consistent with human fetal hair follicles and mouse guard, awl and auchene hair follicles[17]. These data show that the hair follicles of skin organoids undergo a morphogenesis similar to that of mammalian hair follicles. Moreover, the organoid hair follicles are periodically spaced throughout the epidermis, suggesting that appendage patterning mechanisms are preserved in the organoids[18].

Organoids have craniofacial identity

To gain insight into the cell lineages that arise in skin organoids, we performed single-cell RNA sequencing (scRNA-seq) on WA25 and DSP-GFP organoids after one week (day 6; 23,329 cells) and after one month (day 29; 18,190 cells) of differentiation. For both time points, the WA25 and DSP-GFP datasets were integrated to identify differences and similarities in cellular composition (Fig. 2; see Supplementary Data 1–6 for detailed analysis). At day 6, unbiased clustering identified 24 subtypes, which we manually annotated into 4 major cell groups: epithelial (around 59%), neuroglial (around 8%), mesenchymal (around 8%) and actively cycling cells (around 25%) (Fig. 2b, Extended Data Fig. 5). All cell subtypes were present in WA25 and DSP-GFP organoids, but in different proportions (Fig. 2b, c), suggesting that further optimization of the protocol may be needed to reduce variability[19]. The largest group—epithelial cells—consisted of *EPCAM*$^+$*TFAP2A*$^+$ cells that expressed markers of nascent surface ectoderm (*HAND1*$^+$), epidermal progenitors (*WNT6*$^+$*TP63*$^+$), and anterior (*SIX1*$^+$*PAX6*$^+$*DLX5*$^+$) and posterior (*SIX1*$^+$*PAX8*$^+$*GBX2*$^+$) cranial placodes[11,20] (Fig. 2a–d, Extended Data Fig. 5). Within the neuroglial group, *PAX6*$^+$*HES5*$^+$ neuroectoderm-like cells clustered near *PLP1*$^+$*SOX10*$^+$*PAX3*$^+$ CNC-like cells (Fig. 2d), suggesting that CNC cells arise from a neuroectoderm intermediate[11]. The mesenchymal group contained *NR2F1*$^+$*MEF2C*$^+$*TWIST1*$^+$ cells, consistent with CNC-cell-derived mesenchyme that is capable of generating dermal cells[21] (Fig. 2d, Extended Data Fig. 5). No cell clusters expressed early markers of the endoderm or mesoderm lineages, such as *SOX17* or *TBXT*. These data show that skin organoids emerge from anatomically diverse populations of cranial ectoderm cells.

Clustering analysis at day 29 revealed that mesenchymal cells were the dominant cell group (around 70%) over the epithelial (around 17%), neuroglial (around 7%) and actively cycling (around 6%) cell groups (Fig. 2e–h; see Supplementary Data 4–6 for detailed analysis). The epithelial group contained three epidermal sub-clusters: *CXCL14*$^+$ basal, *KRT1*$^+$ intermediate and *KRT4*$^+$ peridermal keratinocytes[22] (Fig. 2e, Extended Data Fig. 5). Any remaining surface ectoderm progenitor cells or placodal cells were undetectable. The neuroglial group contained clusters of Schwann cell precursor-like cells (*PLP1*$^+$*MPZ*$^+$), as well as melanocytes (*PMEL*$^+$*MITF*$^+$) (Fig. 2h, Extended Data Fig. 5). Neural clusters (within the neuroglial group) contained cells that expressed *SOX2* (neural progenitors), *DCX* (neurons of the central nervous system) or *PRPH* (neurons of the peripheral nervous system)[23] (Extended Data Fig. 5). The mesenchymal group contained ten distinct *PRRX1*$^+$ populations, including dermal fibroblast-like subtypes that expressed *TRIL*, *TWIST2*, *APCDD1* and *PAX1*[24] (Fig. 2h, Extended Data Fig. 5). Other mesenchymal clusters contained cells that expressed *DLX* genes, similar to CNC-cell-derived mesenchymal cells of the pharyngeal arches or frontonasal process[13] (Fig. 2h, Extended Data Fig. 5). Notably, these clusters also had cells that expressed *PITX1*, *BARX1*, *GSC* and *HAND2*—markers of the first pharyngeal arch, which gives rise to the jaw and the outer and middle ear during development[25] (Fig. 2h). Furthermore, we noted a lack of expression of HOX genes, which would indicate more-caudal

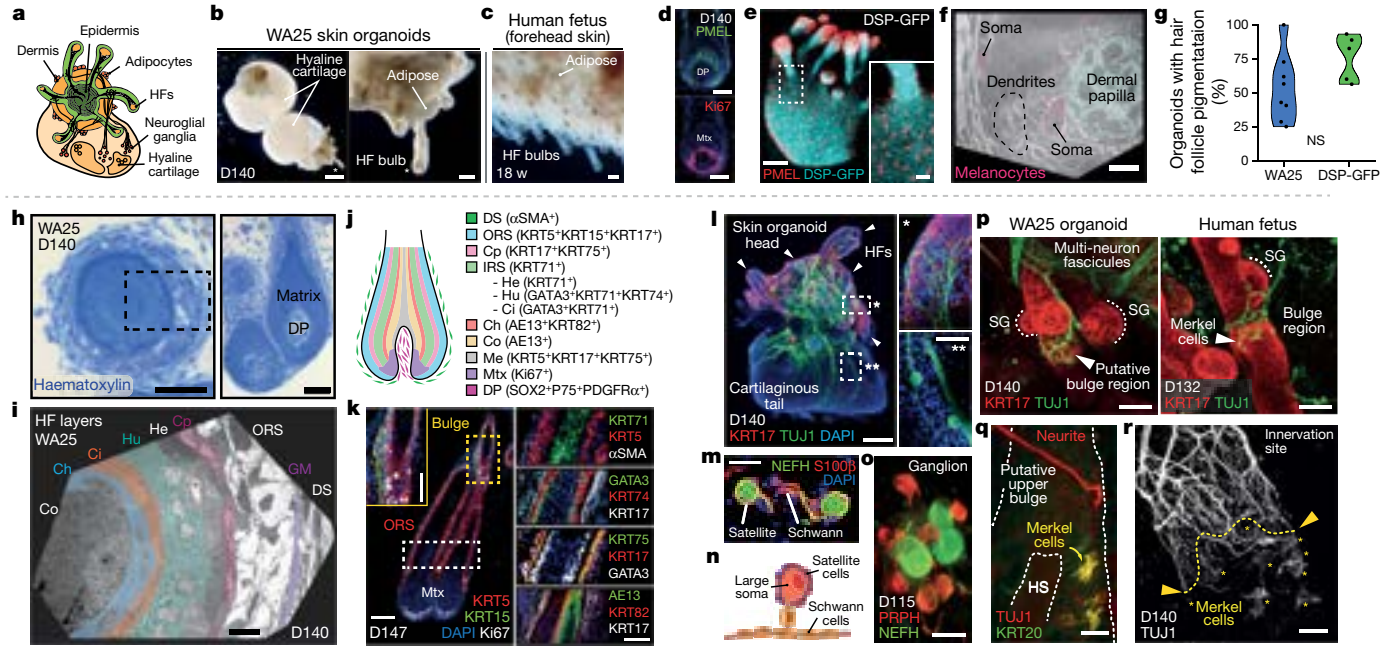


Fig. 3 | Pigmentation and structure of hair follicles and neural innervation of skin organoids. a–**c**, Skin organoid anatomy. **a**, Schematic of a typical skin organoid. **b**, **c**, Dark-field images of day-140 WA25 skin organoids, showing cartilage in the tail (**b**), and human fetal forehead skin at 18 weeks (**c**). The area marked with the asterisk (**b**) is magnified to the right. Scale bars, 250 µm (**b**, left); 100 µm (**b**, right), (**c**). **d**–**g**, Melanocytes. **d**, Melanocytes in WA25 organoid hair follicles immunostained for PMEL and Ki67. **e**, PMEL$^+$ melanocytes in the hair follicles and epithelium of DSP-GFP organoids (Supplementary Video 3). **f**, Transmission electron microscopy (TEM) image showing matrix-associated melanocytes (pink). Scale bars, 100 µm (**e**, main image); 50 µm (**d**); 25 µm (**e**, inset); 10 µm (**f**). **g**, Violin plots comparing the frequency of hair follicle pigmentation between WA25 cultures (n = 137 organoids) and DSP-GFP cultures (n = 91 organoids); Welch's two-sided t-test, P = 0.077 (not significant; NS). **h**–**k**, Hair follicle layers. **h**, Haematoxylin staining in day-140 WA25 hair follicles. **i**, TEM image of the region in the dashed box in **h**, showing hair follicle layers. Ch, cuticle; Ci, inner root sheath (IRS) cuticle; Co, cortex; Cp, companion layer; DS, dermal sheath; GM, glassy membrane; He, Henle's layer; Hu, Huxley's layer; Me, medulla; ORS, outer root sheath. **j**, Schematic of human hair follicle layers. **k**, Immunostaining for markers in **j** in the hair follicles of day-147 organoid, showing each layer from outermost αSMA$^+$ dermal sheath to innermost AE13$^+$ cortex. Scale bars, 100 µm (**k**, left); 50 µm (**h**, right, **k**, right, inset); 25 µm (**h**, left); 10 µm (**i**). **l**–**r**, Neural network. **l**, Whole-mount sample of day-140 WA25 skin organoids, showing TUJ1$^+$ neurons contacting and wrapping KRT17$^+$ epithelium and hair follicles. Dashed boxes are magnified to the right. Asterisks indicate the area in which neurites are located that target the hair follicle bulge (*) or the neuron soma (**). Arrowheads indicate hair follicle clusters. **m**, NEFH$^+$ neurons with S100β$^+$ satellite glial and Schwann cells. **n**, Schematic of typical skin organoid neurons. **o**, Whole-mount sample of a day-115 WA25 skin organoid, showing a ganglion with PRPH$^+$NEFH$^-$ and PRPH^{low}NEFH$^+$ neurons (Supplementary Video 5). **p**, Whole-mount samples of day-140 WA25 skin organoids stained for TUJ1 and KRT17 (left) and human fetal forehead skin at 18 weeks (right). Neurites innervate the upper bulge of the hair follicles. Sebaceous glands (SG) are indicated. **q**, **r**, Whole-mount sample of day-140 WA25 skin organoids. TUJ1$^+$ neurites innervate the upper bulge of the hair follicle above TUJ1$^+$KRT20$^+$ (yellow) Merkel cells. HS, hair shaft. **r**, High-magnification image of the innervation site. Yellow asterisks, Merkel cells. See Supplementary Video 6. Scale bars, 250 µm (**l**, left), 50 µm (**p**), 25 µm (**l**, right, **m**, **o**, **q**, **r**). See 'Statistics and reproducibility' in Methods for plot and experimental information.

pharyngeal arches[26] (Extended Data Fig. 5). These data suggest that skin organoids may develop a distinct anatomical identity akin to that of the skin of the chin, cheek and outer ear, despite an early indication at day 6 of broad anterior–posterior patterning. The mesenchymal group also contained a small subset of *TNNT1*$^+$*MYOG*$^+$ myocyte-like cells, potentially suggesting an off-target induction of mesoderm (Extended Data Fig. 5e, f, Supplementary Data 6). We did not, however, detect progenitors for cells of other lineages—such as Langerhans cells, which are immune cells that reside in the skin.

We also performed scRNA-seq on day-48 WA25 organoids, and observed more-definitive markers of the dermal lineage compared to the day-29 organoids. In particular, five *VIM*$^+$ clusters contained cells that expressed *DPT*, *TWIST2* and *LUM*, as well as several collagen genes, including *COL1A1*, *COL1A2*, *COL3A1*, *COL5A3*, *COL6A3*, *COL12A1* and *COL21A1*[24] (Extended Data Fig. 6a–d; see Supplementary Data 7 for detailed analysis). In addition, we gained insight into self-organizing signalling mechanisms that may govern interactions between the epidermal and dermal layers. Genes encoding modulators of the WNT signalling pathway were differentially expressed between epidermal (*WNT6* and *LEF1*) and dermal (*SFRP2*, *TCF4*, *WIF1* and *APCDD1*) cell populations[27] (Extended Data Fig. 6d, e). Presumptive dermal cells expressed *FGF7* (encoding keratinocyte growth factor), which may be a key driver of epidermal stratification in this system[28,29] (Extended Data Fig. 6f). These data show that skin organoids generate diverse cell populations that are consistent with those of embryonic skin.

Hair follicle pigmentation, layering and innervation

After more than 100 days in culture, skin organoids had the morphology illustrated in Fig. 3a. The external appearance of skin organoids was comparable to 18-week human fetal skin viewed from the dermal side (Fig. 3b, c). Notably, organoid hair follicles were often pigmented, and we thus examined the development of melanocytes in more detail (Fig. 3b, d–g, Extended Data Fig. 7a, c–g). In pigmented skin organoids, we found melanocytes that express premelanosome protein (PMEL) distributed evenly throughout the epithelium and concentrated in the matrix region of hair follicles (Fig. 3d, e, Supplementary Video 3). Electron microscopy confirmed the production of melanosomes in melanocytes of the epidermis and hair follicles (Fig. 3f, Extended Data Fig. 7a, c). An average of 53.5% WA25 and 76.2% DSP-GFP organoids produced pigmented hair follicles (Fig. 3g). Using immunostaining and electron microscopy, we found that the organoid hair follicles had most of the unique cellular layers of hair follicles—except the medulla layer, which is characteristic of adult terminal hair follicles. This suggests that the

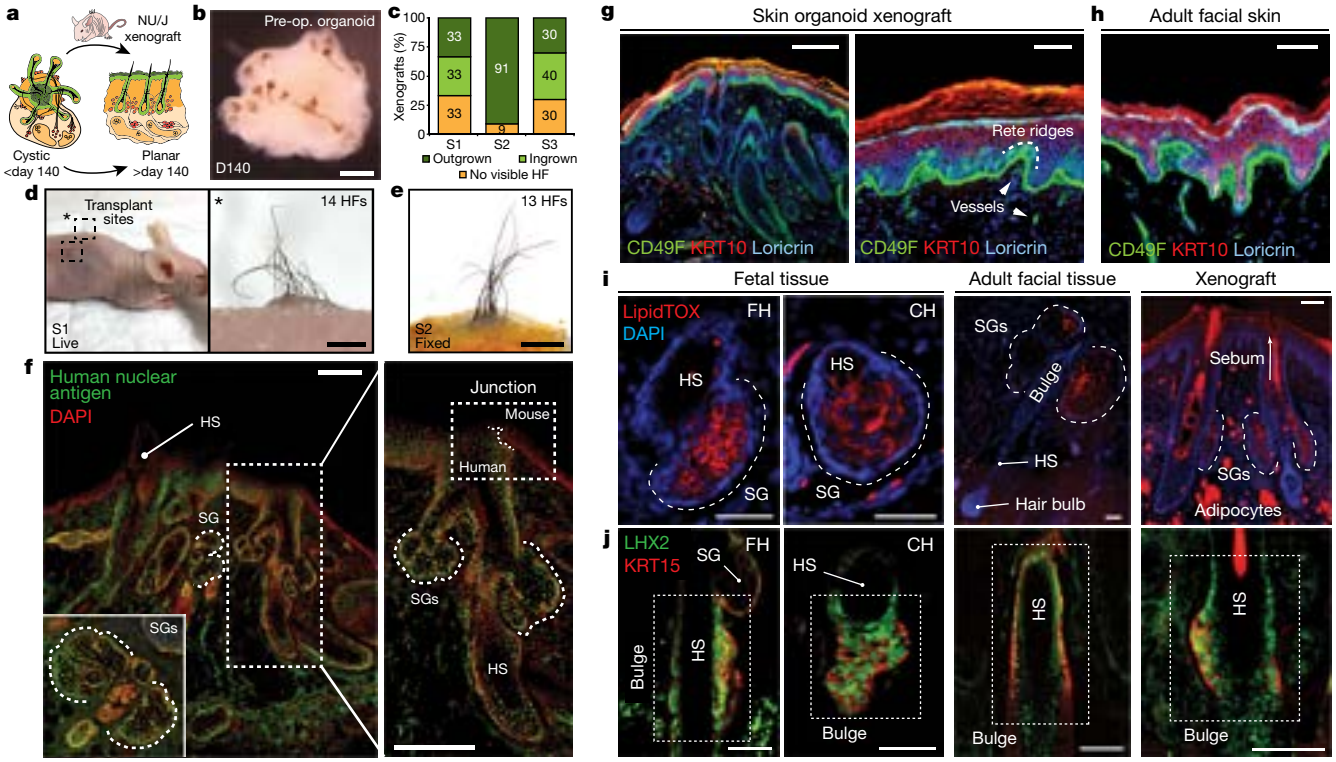


Fig. 4 | Skin organoids undergo cystic-to-planar transition when xenografted. a, Schematic of skin organoid xenografting strategy. **b**, Dark-field image of a day-140 WA25 skin organoid before grafting (pre-operative; pre-op.). Scale bar, 200 µm. **c**, Quantification of hair follicle growth percentage throughout xenograft experiments. Data are compiled from 27 xenografts performed over 3 separate experiments (surgeries S1, S2 and S3). **d**, A xenografted NU/J mouse at 38 days after S1 (skin organoids are 178 days old in total). Pigmented hair shafts (dashed boxes) are visible at both graft sites. One graft site (right, asterisk) shows 14 hair follicles. Scale bar, 1 mm. **e**, S2 xenograft containing pigmented hairs at 49 days after S2 (skin organoids are 189 days old in total). Scale bar, 1 mm. **f**, Immunostaining for human nuclear antigen and DAPI in a xenograft. Note the junction between mouse and human epidermis (right, dashed box). Organoid hair follicles have sebaceous glands. Scale bars, 200 µm. **g**, **h**, Comparison of epidermal layers between skin organoid xenograft (**g**) and adult facial skin (**h**) by immunostaining for CD49F (basal layer), KRT10 (intermediate layer) and loricrin (granular layer). Unspecific staining above the loricrin-positive granular layer is the cornified layer. Note the Rete ridges in xenograft and adult skin. Arrowheads indicate vessels. Scale bars, 200 µm (**g**, left); 50 µm (**g**, right, **h**). **i**, **j**, Comparison of LipidTOX$^+$ sebaceous glands (**i**) and LHX2$^+$KRT15$^+$ putative bulge stem cells (**j**) at the hair follicle bulge region in fetal forehead and cheek skin (left), adult facial skin (middle) and skin organoid xenograft (right). LipidTOX also labels adipocytes in **i**. Dashed lines in **i** indicate sebaceous glands. Dashed boxes in **j** indicate hair follicle bulges. CH, cheek; FH, forehead. Scale bars (**i**, **j**), 100 µm (adult and xenograft); 50 µm (fetal).

skin organoids produce hair follicles similar to those of lanugo or vellus hairs[30] (Fig. 3h–k; see Supplementary Discussion on hair follicle type). Another feature of late-stage organoids was hyaline cartilage, which developed in the tail region (Fig. 3a, b, Extended Data Fig. 8). Chondral progenitor cells that expressed *ACAN* and *COL9A1*, *COL9A2* and *COL9A3* were evident in our day-48 scRNA-seq dataset (Extended Data Fig. 8). In addition, lipid-rich adipocytes developed around organoid hair follicles, mimicking the development of hypodermal fat[14] (Fig. 3b, c, Extended Data Fig. 9). These data provide evidence for a model of tissue architecture in which skin organoids represent craniofacial skin (superficial tissue) and cartilage (deep tissue) developing 'inside-out'.

We next investigated whether the neuroglial cells identified by scRNA-seq at day 29 matured into organized nervous tissue. In 100% of the organoids examined (more than 30 organoids; 7 experiments), we found a network of neurons that expressed ISL1 and tubulin β class III (TUJ1) with axons interwoven between hair follicles (Fig. 3l, p–r, Extended Data Fig. 7h, Supplementary Videos 2, 4–6). The organoid neurons were reminiscent of neural-crest-derived neurons that are found in cranial or dorsal root ganglia, with medium-to-large-soma neurons that expressed neurofilament heavy chain (NEFH) and peripherin (PRPH), and NEFH$^-$PRPH$^+$ neurons with a smaller soma[23,31] (Fig. 3o, Supplementary Video 7). The most common neuronal morphology was consistent with NEFH$^+$ pseudounipolar neurons, which are similar to Aβ or Aδ-field low-threshold mechanoreceptors or proprioceptors[31] (Fig. 3m, n). These neurons formed fascicles that were associated with S100β$^+$ Schwann-like cells and satellite glia-like cells covering the neuron cell bodies (Fig. 3m, Extended Data Fig. 7i). In large ganglia (more than 10 neurons), we identified a subset of neuron somas that expressed parvalbumin (PVALB)—indicative of proprioceptors[31] (Extended Data Fig. 7j). In organoids at day 125 or later, neuronal processes contacted the epithelium and wrapped around hair follicles, similar to the immature circumferential and lanceolate nerve endings of hair follicles in the human fetus at 18 weeks (Fig. 3l, p–r, Supplementary Videos 5, 6). KRT20$^+$TUJ1$^+$ Merkel cells were located in the outer root sheath of hair follicles near the site of axon targeting, which we presume to be the bulge region of the hair follicles (Fig. 3q, r, Supplementary Video 5). Merkel-like cells expressing key marker genes—for example, *ATOH1*, *KRT8*, *KRT18*, *SOX2* and *ISL1*—were also identified in our day-48 scRNA-seq dataset[32] (Extended Data Fig. 6g). Our data suggest that skin organoids form nascent mechanosensory complexes[33,34]. Further molecular and physiological profiling will be necessary to determine whether other somatosensory neuron subtypes and nerve endings can be induced in skin organoids.

Organoids form planar skin in vivo

After around 150 days in culture, we observed a build-up of squamous cells in the core of skin organoids (Extended Data Fig. 9b) and abnormal

hair follicle morphologies, suggesting that this time point may be the upper limit for skin organoid culture. Thus, we tested whether skin organoids could integrate into endogenous skin in a mouse model (Fig. 4a). On day 140, we removed the cartilaginous tails with tungsten needles (Fig. 4b) and implanted hair-bearing WA25 skin organoids in small 1–2-mm incisions in the back skin of nude mice[35]. Notably, we observed 2–5-mm hairs growing out from 55% of the xenografts (15 of 27 xenografts, 3 experiments) (Fig. 4c–e). The other grafts had ingrown hairs (22%) or failed (22%) owing to technical issues (Fig. 4c). None of the grafts showed ulceration or other signs of uncontrolled tumour-like growth. For xenografts with hair outgrowth, histology confirmed that the organoid epidermis had integrated with the host epidermis, and the hair shafts were oriented perpendicular to the skin surface, similar to bona fide human skin tissue (Fig. 4f, g). We noted that the epidermis of xenografts with hair outgrowth had a cornified layer and Rete-ridge-like structures comparable to adult facial skin (Fig. 4g, h). In addition, $CD49F^+$ vasculature had grown into the xenografts (Fig. 4g). Our findings demonstrate that cystic skin organoids can unfurl and integrate into planar skin at a wound site.

Finally, we examined xenografted hair follicles for two hallmarks of mature pilosebaceous units: sebaceous glands and bulge stem cells. We detected sebaceous glands containing $SCD1^+$ cells in all xenografted hair follicles (Fig. 4i, Extended Data Fig. 9d). Organoid sebocytes had sebum vacuoles that were visible using electron microscopy (Extended Data Fig. 9c). Morphologically, the sebaceous glands of xenografts had multiple lobes, reminiscent of adult sebaceous glands, and were located superficial to the bulge region of the hair follicle (Fig. 4i, Extended Data Fig. 9). In the presumptive bulge region of xenografted hair follicles, we found $KRT15^+LHX2^+NFATC1^+$ cells, similar to hair follicle stem cells, and nephronectin $(NPNT)^+$ basement membrane (Fig. 4j, Extended Data Fig. 10d, Supplementary Video 8). The expression pattern of NFATC1 was predominantly cytoplasmic, which is consistent with follicular bulges in fetal skin; by contrast, adult bulge cells show nuclear NFATC1 expression, a sign of maturity[2,36] (Extended Data Fig. 10). Together, these findings suggest that the hair follicles of skin organoids reach a level of maturity that is roughly equivalent to second-trimester fetal hair and have the cellular components required for further maturation. Long-term observation (longer than a year) will be needed to determine whether xenografted follicles convert to a terminal hair fate or maintain a growth cycle in vivo.

Discussion

Our study establishes a model for investigating the cellular dynamics of developing human skin and its appendages—including sweat glands, which were absent from the skin organoids reported here. A range of genetic skin disorders and cancers could be modelled with skin organoids to accelerate drug discovery. Moreover, with additional control of off-target cell lineages, skin organoids could be used to reconstitute appendage-bearing skin in patients with skin burns or wounds.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-020-2352-3.

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Methods

Human pluripotent stem cell lines and culture

Culture experiments were performed with the WA25 human embryonic stem cell line (passages 21–47), purchased from WiCell Research Institute, and the desmoplakin-mEGFP (DSP-GFP) human induced pluripotent stem cell line (passages 35–51), acquired from the Allen Institute for Cell Science and the Coriell Institute[37]. The cell lines tested negative for mycoplasma prior to experimentation. Cells were cultured on 6-well plates coated with vitronectin recombinant human protein (Invitrogen) at a concentration of 0.5 μg cm^{-2}. Pluripotent stem cells were maintained in Essential 8 Flex (Gibco) medium with 100 μg ml^{-1} Normocin (Invivogen; hereafter, E8). The medium was replenished every other day or every day depending on cell confluency. Cells were passaged at around 80% confluency (generally every 4 to 5 days): WA25 cells were passaged in tiny clusters with 0.5 mM EDTA, and DSP-GFP cells were passaged as single cells (occasionally in tiny clusters) with StemPro Accutase Cell Dissociation Reagent (hereafter, Accutase; Gibco). Either Accutase-exposed or single cells (DSP-GFP) were passaged in E8 medium containing 10 μM Y27632 (a ROCK inhibitor that inhibits apoptosis; Stemgent). Twenty-four hours after passaging, the medium was replenished with fresh E8 medium without Y27632. Detailed information about cell line specifics, validation and testing is available at https://www.wicell.org/home/stem-cells/catalog-of-stem-cell-lines/wa25.cmsx?closable=true and https://www.allencell.org/cell-catalog.html.

Optimized differentiation of human pluripotent stem cells

For differentiation, colonies of human pluripotent stem cells were detached from the culture dish using Accutase. Once dissociated, cells were collected as a single-cell suspension in E8 medium containing 10 μM Y27632 (hereafter, E8-10Y). Cell concentration was determined with an automated cell counter (Countess II Automated Cell Counter; Life Technologies); Trypan blue (Gibco) was used at a ratio of 1:1 (Trypan blue:cell suspension) to distinguish dead and live cells. The appropriate number of live cells needed for an experiment was transferred into E8 medium containing 20 μM Y27632 (hereafter, E8-20Y). From the E8-20Y cell suspension, cells were distributed at a count of 3,500 cells in 100 μl per well into 96-well U-bottom plates (Thermo Fisher Scientific). Aggregation was aided by centrifugation at 110*g* for 6 min. These cell aggregates were incubated in a 37 °C incubator under 5% CO_2 for 48 h. Thus, we designated this time point as day −2. After 24-h incubation (day −1), fresh 100 μl E8 medium was added to each well to dilute out Y27632 and promote cell proliferation and aggregate growth. On day 0, to start differentiation, all cell aggregates were individually collected and transferred to new 96-well U-bottom plates in 100 μl of E6-based differentiation medium containing 2% Matrigel (Corning), 10 μM SB431542 (Stemgent), 4 ng ml^{-1} basic FGF (hereafter, FGF; PeproTech) and 2.5 ng ml^{-1} BMP4 (PeproTech) to initiate non-neural ectoderm formation. On day 3 of differentiation, to induce CNC cell formation, 200 ng ml^{-1} LDN (BMP inhibitor, Stemgent) and 50 μg ml^{-1} FGF were added in a volume of 25 μl per well, thus making the final volume 125 μl per well. On day 6 of differentiation, 75 μl of fresh E6 medium was added, bringing the final volume to 200 μl. Half of the medium was changed (removal of 100 μl spent medium and addition of 100 μl fresh E6 medium) on days 8 and 10. On day 12, to induce self-assembly of the epidermis, all aggregates were transferred into individual wells of 24-well low-attachment plates (Thermo Fisher Scientific) in 500 μl organoid maturation medium (OMM) containing 1% Matrigel. To maintain the aggregates in a floating culture for constant medium circulation, 24-well plates were placed on an orbital shaker shaking at 65 rpm in a 37 °C incubator with 5% CO_2. OMM is composed of advanced Dulbecco's modified Eagle medium: nutrient mixture F-12 (DMEM/F-12; Gibco) and neurobasal medium (Gibco) at a 1:1 ratio, 1× GlutaMax (Gibco), 0.5× B-27 minus vitamin A (Gibco) and 0.5× N-2 (Gibco) supplements, 0.1 mM 2-mercaptoethanol (Gibco) and 100 μg ml^{-1} normocin. On differentiation day 15, half of the spent medium was replenished (removal of 250 μl of spent medium and addition of 250 μl fresh medium) with OMM containing 1% Matrigel. Starting from day 18, half of the medium was replenished every three days (from day 18 to day 45) or every other day (from day 45 to day 150 or longer) with fresh OMM without Matrigel, including a full medium change once a week starting from day 45. Increasing the total volume of medium per well to 1 ml was at times necessary from day 80 onward as aggregates mature and grow larger. (See Supplementary Note for more information on the differentiation protocol and Supplementary Table 2 for the compositions of the media). A more detailed step-by-step skin organoid induction protocol can be found on the Protocol Exchange[38].

Fetal and adult human tissue specimens

Facial skin tissue from an 18-week miscarried fetus was obtained from the University of Washington Birth Defects Research Laboratory. The tissue was not obtained from a living individual and was de-identified. De-identified samples of adult human facial skin tissue were obtained from the Human Skin Disease Resource Centre at the Brigham and Women's Hospital, Harvard Medical School. Informed consent was obtained from all patients (or their legal guardians) who contributed samples used in this study. All specimens qualified for NIH Exemption 4 and were deemed to be 'Non-Human Subjects Research' by the Indiana University School of Medicine Institutional Review Board.

Skin organoid dissociation for scRNA-seq

To dissociate WA25 and DSP-GFP cell aggregates into single cells, randomly selected day-6 organoids ($n = 10$) and day-29 organoids ($n = 5$) were pooled from each group. For day-48 WA25 cell aggregates, six representative skin organoids were collected, and the tail structure containing non-skin-associated mesenchymal and neuronal cells was removed using two tungsten needles before dissociation to bias analysis towards skin tissue. In brief, collected organoids were incubated with pre-warmed (37 °C) TrypLE for a total of 8–30 min (depending on the organoid sizes and densities) in a 37 °C incubator on a shaker at 65 rpm for a gentle swirl. During the incubation, the dissociation mixture was agitated every 3 to 10 min with gentle pipetting using wide-orifice p1000 and p200 tips (Mettler-Toledo). By 8–30 min of incubation, the tissue structure dissociated into a single-cell suspension with no visible cell aggregation. Cold 3% bovine serum albumin (BSA) solution (Sigma-Aldrich) was added to the dissociated cell suspension to inactivate TrypLE enzymatic activity. The suspension was filtered through a 40-μm Flowmi cell strainer (Bel-Art) to eliminate any debris. An additional three washes with cold 3% BSA solution were performed. Cells were resuspended in 3% BSA and filtered through a 40-μm Flowmi cell strainer. Viability (live cell percentage) and live cell count was determined using Trypan blue (used at a 1:1 ratio) and Countess II. The final cell concentration was around 1,000 cells μl^{-1}, with a cell viability greater than 90.

scRNA-seq cDNA library preparation and sequencing

Single-cell 3′ RNA-seq experiments were conducted using the Chromium single cell system (10X Genomics) and the NexSeq500 or NovaSeq 6000 sequencer (Illumina). Approximately 17,000 cells (targeting 10,000 cells) per sample were added to a single-cell master mix, following the Chromium Single Cell 3′ Reagent Kits v2 User Guide, CG00052 Ver B (10X Genomics). Day-6 samples were processed using the v3 Chromium Single Cell 3′ Reagent Kits. Along with the single-cell gel beads and oil partitioned in separate wells of a Single Cell A Chip, the single-cell reaction mixture was loaded to the Chromium Controller for Gel Bead-in-Emulsion (GEM) generation and barcoding, followed by cDNA synthesis and library preparation. At each step, the quality of cDNA and library was examined by Bioanalyzer. The resulting library was sequenced in a custom program for 26-bp plus 98-bp paired-end sequencing on an Illumina NextSeq500 or 28-bp plus 91-bp paired-end

sequencing on an Illumina NovaSeq 6000 to a read depth of more than 30,000 reads per cell.

scRNA-seq data analysis

The 10X Genomics Cell Ranger 2.1.0 pipeline (http://support.10xgenomics.com/) was used to process raw sequence data. In brief, Cell Ranger uses bcl2fastq (https://support.illumina.com/) to demultiplex raw base sequence calls generated from the sequencer into sample-specific FASTQ files. The FASTQ files are then aligned to the reference genome with RNA-seq aligner STAR. The aligned reads are traced back to the individual cells, and the gene expression level of individual genes is quantified on the basis of the number of unique molecular indices (UMIs) detected in each cell. Filtered gene–cell barcode matrices were generated by Cell Ranger for further analysis. Cells with an extremely high or low number of detected UMIs were excluded from further analysis. In addition, cells with a high percentage of mitochondrial reads were filtered out. After removing unwanted cells, the gene expression levels for each cell were normalized by the total number of UMIs in the cell and multiplied by a scaling factor of 10,000. After log-transformation, we used Seurat v.2 for cell clustering using principal component analysis (PCA) on highly variable genes[39,40]. The cell clusters were visualized using the *t*-distributed stochastic neighbour embedding (*t*-SNE) and UMAP plotting methods. The gene markers for each cluster were identified through differential expression analysis by comparing cells in the cluster to all other cells.

To integrate the single-cell data from WA25 and DSP-GFP samples, we applied functions FindIntegrationAnchors and IntegrateData from Seurat v.3[41]. The integrated data were then scaled and PCA was performed. Clusters were identified with the Seurat functions FindNeighboursand FindClusters. The FindConservedMarkers function was subsequently used to identify canonical cell-type marker genes that are conserved in cells across different samples. Cell cluster identities were manually defined with the cluster-specific marker genes. To compare average gene expression within the same cluster between cells of different samples, we applied the function AverageExpression. The cell clusters were visualized using *t*-SNE and UMAP plots. The R packages ggplot2 and ggrepel (https://github.com/slowkow/ggrepel) were used to plot the average gene expression. Violin plots (VlnPlot) and feature plots (FeaturePlot) were used to visualize specific gene expression across clusters and different sample conditions.

Immunohistochemistry

For immunostaining, fixed samples were cryoprotected through a graded treatment process of 10%, 20% and 30% sucrose (Sigma-Aldrich), embedded on cryomolds (Endwin Scientific) in tissue-freezing medium (General Data Healthcare), snap-frozen at −80 °C and sliced to a thickness of 12 or 15 µl. Cryosections were blocked in 10% normal goat or horse serum, incubated with primary antibodies diluted in 3% normal goat or horse serum and then incubated with secondary antibodies in 3% normal goat or horse serum. Unless stated otherwise, images are representative of specimens obtained from at least three separate experiments. For immunohistochemistry (IHC) analysis of aggregates between days 0 and 12, we sectioned 3–6 aggregates from each condition in each experiment. IHC analysis of later stages of development was performed on at least three aggregates from each condition per experiment.

Whole-mount immunostaining was performed using a previously published method[42] with major adjustments in incubation times to account for reduced tissue size[2]. In brief, samples were fixed in 4% (v/v) paraformaldehyde (PFA; Electron Microscopy Sciences), permeabilized, descaled, incubated with primary antibodies, incubated with fluorescently labelled secondary antibodies, re-fixed in 4% (v/v) PFA (Electron Microscopy Sciences) and cleared. Microscopy was performed with a Leica DMi8 Inverted Microscope, a Leica TCS SP8 Confocal Microscope, an Olympus FV1000-MPE Confocal/Multiphoton Microscope and a Nikon A1R HD25 Microscope. Three-dimensional reconstruction was performed with the Imaris 8 software package (Bitplane) on computers housed at the Indiana Centre for Biological Microscopy and NIS-Elements analyser software provided from Nikon. See Supplementary Table 3 for a list of antibodies.

Cell culture quantitative analysis

Frequency of organoids producing hair follicles. Starting from day 75, organoids with protruding hair placodes, germs, pegs and hair follicles were counted and recorded. To calculate a percentage of hair follicle production within a culture, the number of organoids that produced hair follicles was divided by the total number of organoids cultured in the experiment and multiplied by a hundred. Percentages of hair follicle formation from nine independent experiments of both WA25 and DSP-GFP cell lines were calculated and averaged to get the general frequency of hair follicle production from each cell line.

Number of hair follicles produced. The numbers of hair follicles generated in both WA25 and DSP-GFP skin organoids were counted manually. From each cell-line-derived hairy skin organoid in 9 independent experiments, 80 organoids were randomly pooled, and their pictures were taken under a Leica M165FC stereo microscope. An angle of each organoid was chosen blinded, and a picture of that view under the microscope was taken. The number of hair follicles in the image view was quantified by counting the protruded hair bulbs using the ImageJ 'Cell Counter' plug-in function.

Frequency of hair follicle pigmentation. Among the hair-bearing skin organoids, some hair follicles were pigmented in dark brown or black in the bulbs and hair shafts, whereas the rest were albino (white hairs). Pigmentation is visible with the naked eye, but to clearly distinguish pigmented hairs—including faint, pigmentation-initiating hair follicles from white hairs—organoids were observed under a Leica DMi1 microscope without condenser on and/or under a Leica M165FC stereo microscope set to dark-field illumination. The number of skin organoids containing pigmented hair follicles was divided by the total number of hair-follicle-generating organoids within the experiment and multiplied by a hundred to calculate the percentage of pigmented hair follicle formation. The percentages from eight independent WA25 experiments and five independent DSP-GFP experiments were averaged to determine the rate of hair follicle pigmentation for each cell line.

LipidTOX staining

LipidTOX Staining (Invitrogen) was performed to visualize sebaceous glands and lipid-rich adipocytes. Cryosections of skin organoids were incubated with LipidTOX neutral lipid stain diluted at a ratio of 1:200 in 1× PBS for 30 min at room temperature, followed by Hoechst staining (1:2,000; Invitrogen) for 1 min at room temperature to visualize nuclei.

Xenograft experiments

In vivo experiments were performed after approval from the Indiana University Animal Care and Use Committee (IACUC) and complied with federal regulations. For grafting, 5-6-week-old female immunodeficient nude mice (NU/J) purchased from The Jackson Laboratory were used. Before surgery, WA25-cell-derived skin organoids bearing either pigmented or albino hair at various days (days 113, 115, 120, 124, 127, 140 and 145) were pooled and prepared; using tungsten needles, the tail portion (mesenchymal cells or cartilage) of the organoids was gently removed, and a tiny incision was made on the hairy skin to increase the chance of the organoid core being exposed and to allow hairs to grow outward. The mice were anaesthetized under isoflurane, and bupiviczaine, a pain reliever, was administered (6 mg kg^{-1}) by subcutaneous injection to the intended incision sites on the back of the mice (3–5 incision sites per mouse). Incisions of about 1–2 mm (a size that can fit a hairy skin organoid) were made, and each hairy skin organoid prepared ahead

was placed on an individual incision site with the hair bulb side facing inward, contacting the muscle layer of the mouse, and the organoid incision side outward, exposed to the air. Sterile gauze was placed on top of the back of the mouse, covering all graft sites, and tightened with bandage by wrapping around the body of the mouse. Then, the bandage was sutured for stability. After the surgery, the mice were provided with Carprofen wet food for pain relief. The dressing was removed 7–9 days after surgery. The mice were observed for about 7 or 14 weeks after the surgery, and then euthanized. Using a scalpel, each area with skin organoid graft was collected at about 3 mm × 5 mm (width × length) size. Collected samples were flattened on a filter paper to inhibit the curling effect of the skin edges and were fixed in 4% PFA overnight at 4 °C with constant gentle agitation.

Analysis of hair follicle growth in xenografts

For transplantation, skin organoids containing hairs were selected and grafted between days 113 and 145 of differentiation. Hair follicle protrusion and growth at the grafted sites on the back of mice were monitored until the day mice were euthanized. Some hair follicles did not grow outwards, potentially owing to an imperfect organoid grafting method in which the follicle orientation was altered (hair bulbs of organoid facing outward) during the procedure, so that the hairs grew inward instead and were visible as pigmented dark shade under the skin. The number of grafted sites with protruding hairs (both pigmented and albino) or dark pigmented shade under the skin were counted and divided by the number of initial grafts that were made to calculate the percentage of outgrown, ingrown or failed grafts.

Statistics and reproducibility

No statistical methods were used to predetermine sample size and the investigators were not blinded to the treatment groups. In general, experiments were not randomized; however, the organoid specimens used for histology and scRNA-seq analysis were selected at random and not screened for quality or specific morphological characteristics. All statistics were performed using GraphPad Prism 8 software. Unless stated otherwise, data are reported as mean ± s.e.m. Violin plot limits show maxima and minima, and the dots represent individual data points.

Violin plots in Fig. 1g represent the frequency of hair follicle generation in WA25, DSP-GFP and WA01 cultures. WA25: average 87.4 ± 3.5%, minimum 68.8%, maximum 100%, 181 hairy organoids out of a total of 212 skin organoids across 9 independent experiments; DSP-GFP: average 87.2 ± 4.1%, minimum 66.7%, maximum 100%, 177 hairy organoids out of a total of 212 skin organoids across 9 independent experiments; WA01: 70.8%, 92 hairy organoids out of a total of 130 skin organoids in 1 experiment (Supplementary Table 1). Violin plots in Fig. 1h show the average number of hair follicles produced within individual skin organoids across 80 specimens that were collected from 9 independent experiments between days 75 and 147 from WA25 and DSP-GFP cultures. WA25: average 64 hair follicles per organoid, minimum 9, maximum 285; DSP-GFP: average 48 hair follicles per organoid, minimum 7, maximum 128. Violin plots in Fig. 3g compare the frequency of formation of organoids that bear pigmented hairs between WA25 and DSP-GFP cultures. WA25: average 53.5 ± 8.8%, minimum 25%, maximum 100%, 70 pigmented hairy organoids out of a total of 137 hairy organoids across 8 independent experiments; DSP-GFP: average 76.2 ± 7.6%, minimum 56.3%, maximum 93.3%, 72 pigmented hairy organoids out of a total of 91 hairy organoids across 5 independent experiments. Statistical significance was determined in Fig. 3g using an unpaired Student's *t*-test with Welch's correction to account for unequal standard deviations and sample sizes, $P = 0.077$. See 'Cell culture quantitative analysis' in Methods for more information.

All representative skin organoid images shown in this Article represent results from nine independent experiments of WA25 and DSP-GFP cultures and one WA01 culture (Figs. 1c–f, i–l, 3b, d–f, h, i, k–m, o–r). The image of an organoid with pigmented hair shown in Fig. 4b represents 1 of the 27 pre-operative skin organoids used in 3 independent xenograft experiments, and Fig. 4d–j show results from the same experiments. All immunostaining images shown in this Article are representative of a minimum of three replicates with the same staining. Specifically, Figs. 1d, i–l, 3k–m, p–r represent five independent stainings on three to five different organoids per staining, and Fig. 3d, e, o represent three independent stainings on three to five different organoids per staining. The images in Fig. 3f, h, i are representative hair follicle TEM images (Fig. 3f, i) and haematoxylin staining images (Fig. 3h) of two separate skin organoids at day 140.

The culture method was optimized and replicated by three investigators (K.R.K., J.L. and M.S.) using the WA25 and DSP-GFP cell lines. During preparation of the manuscript, K.R.K. and J.L. contacted S.H. and B.M.W. for assistance in validating the method used for skin organoid culture. S.H. and B.M.W. were provided with a detailed protocol for skin organoid induction with a list of suggested reagents. S.H. and B.M.W. were blinded to the data already generated for this study with WA25 and DSP-GFP pluripotent stem cells; thus, they were unfamiliar with the expected morphological changes during skin organogenesis. K.R.K. and J.L. provided minimal guidance via email. In his validation experiment, B.M.W. generated 92 hair-bearing skin organoids (out of a total of 130 organoids) from WA01 embryonic stem cells (Extended Data Fig. 3c).

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this paper.

Data availability

The scRNA-seq data have been uploaded to the Gene Expression Omnibus with accession code GSE147206. In the Supplementary Data, we have provided HTML files that generate an online interface with which to explore our scRNA-seq analysis pipeline and evaluate additional cell cluster markers for the integrated and non-integrated datasets (Supplementary Data 1–7). In addition, the integrated datasets shown in Fig. 2 can be queried via a data exploration portal hosted on the Koehler laboratory website: https://www.koehler-lab.org/resources. Source data for Figs. 1–4 are provided with the Article.

Code availability

Scripts used for single-cell RNA-sequencing analysis are available at https://github.com/Koehler-Lab/Lee_et_al_2020_Nature.

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Author contributions J.L. and K.R.K. conceived the study and wrote the manuscript. J.L. performed most of the in vitro experiments and IHC analysis. J.L., C.C.R., Z.E.P., T.Z.S. and K.R.K. designed and performed the in vivo experiments. M.R.S. and A.K. performed IHC and scRNA-seq data analysis. H.G. and Y.L. performed bioinformatic analyses of scRNA-seq data and generated figures. B.M.W. and S.H. performed protocol validation experiments and generated figures. K.R.K. supervised the project, monitored the experiments and acquired funding. All authors read and approved the final manuscript.

Competing interests J.L. and K.R.K, with the Indiana University Research and Technology Corporation, are inventors on a published patent covering the entire skin organoid induction method and including much of the data presented in this Article (WO2017070506A1). The other authors declare no competing interests.

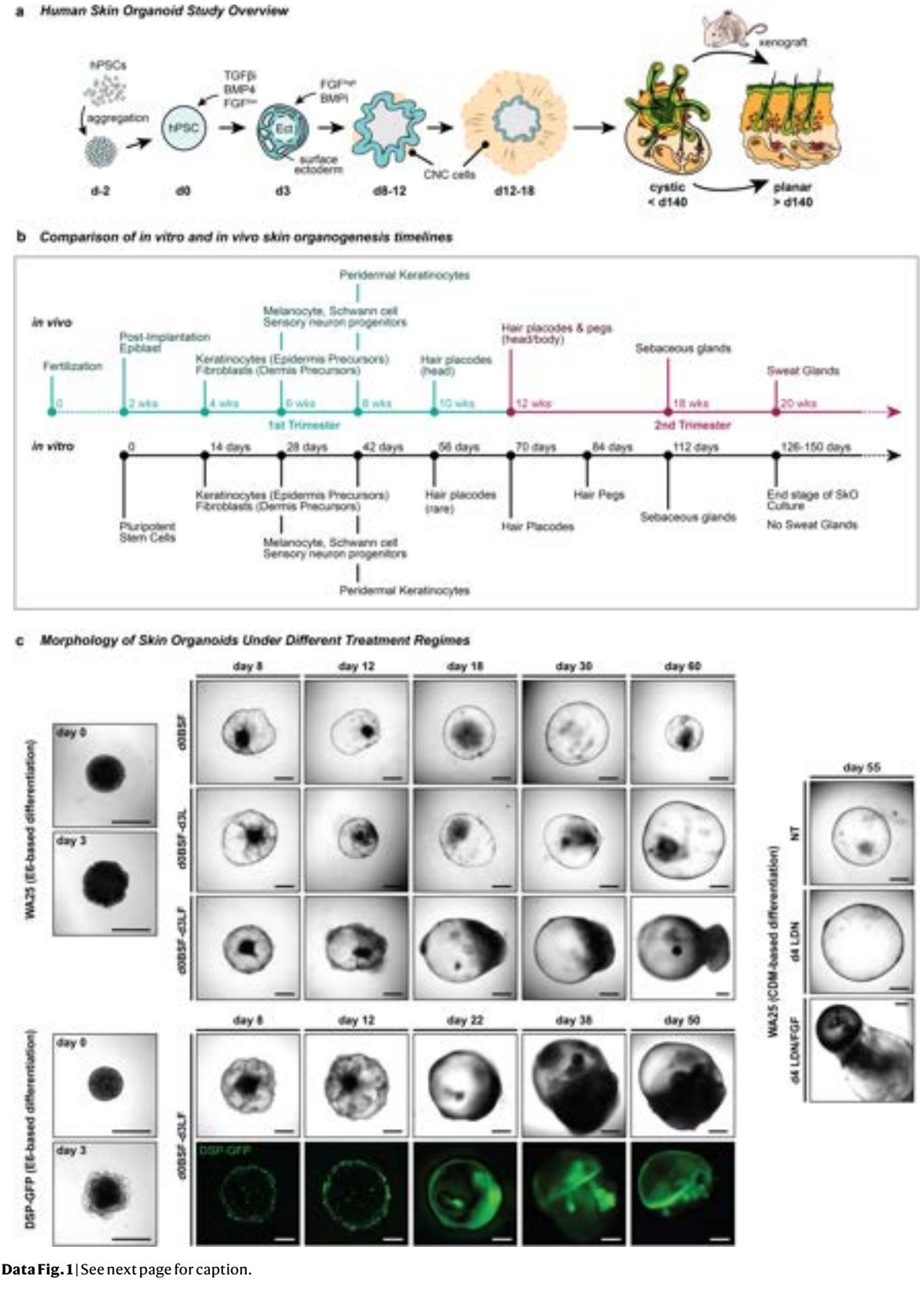
Additional information

Supplementary information is available for this paper at https://doi.org/10.1038/s41586-020-2352-3.

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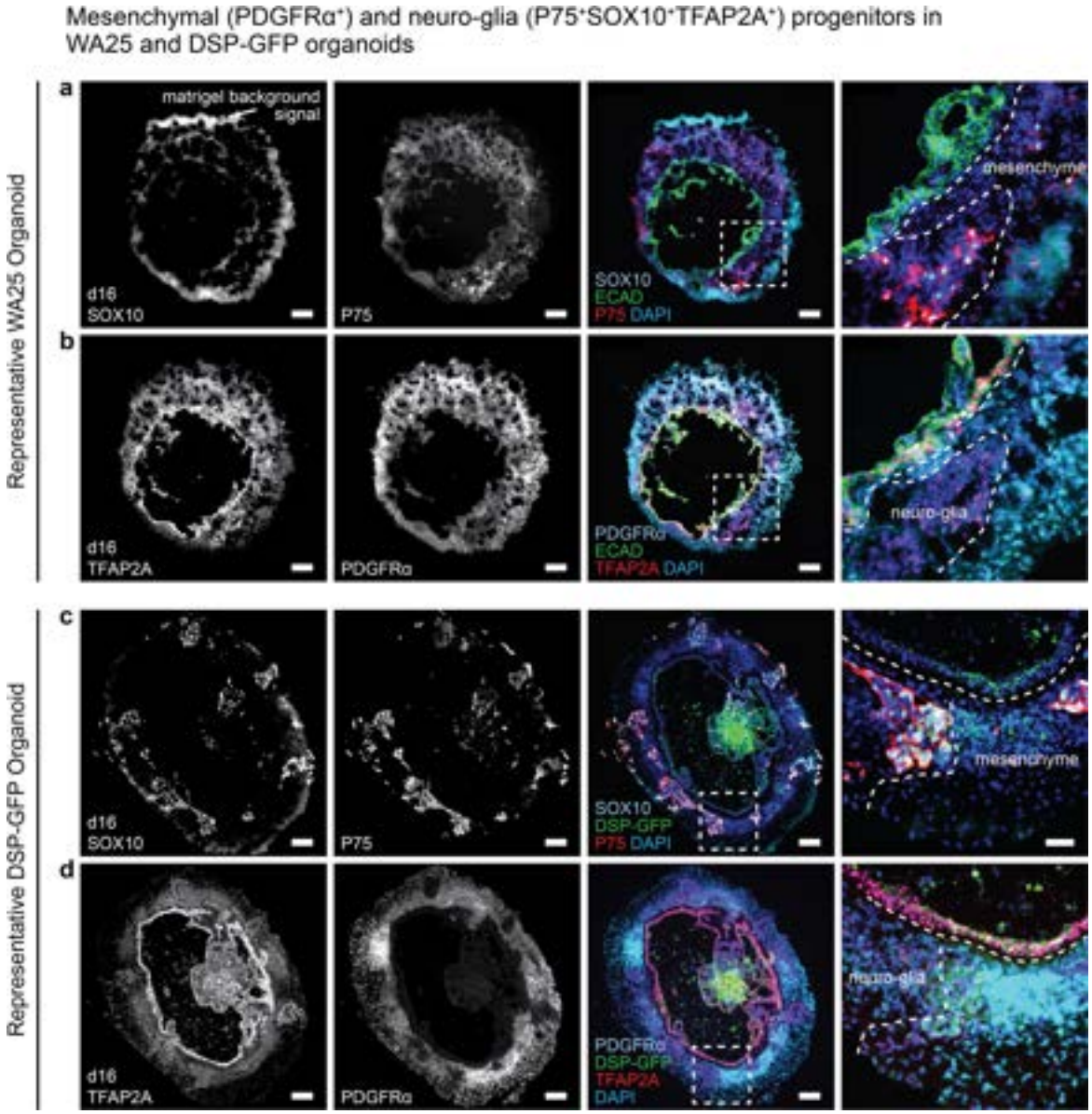
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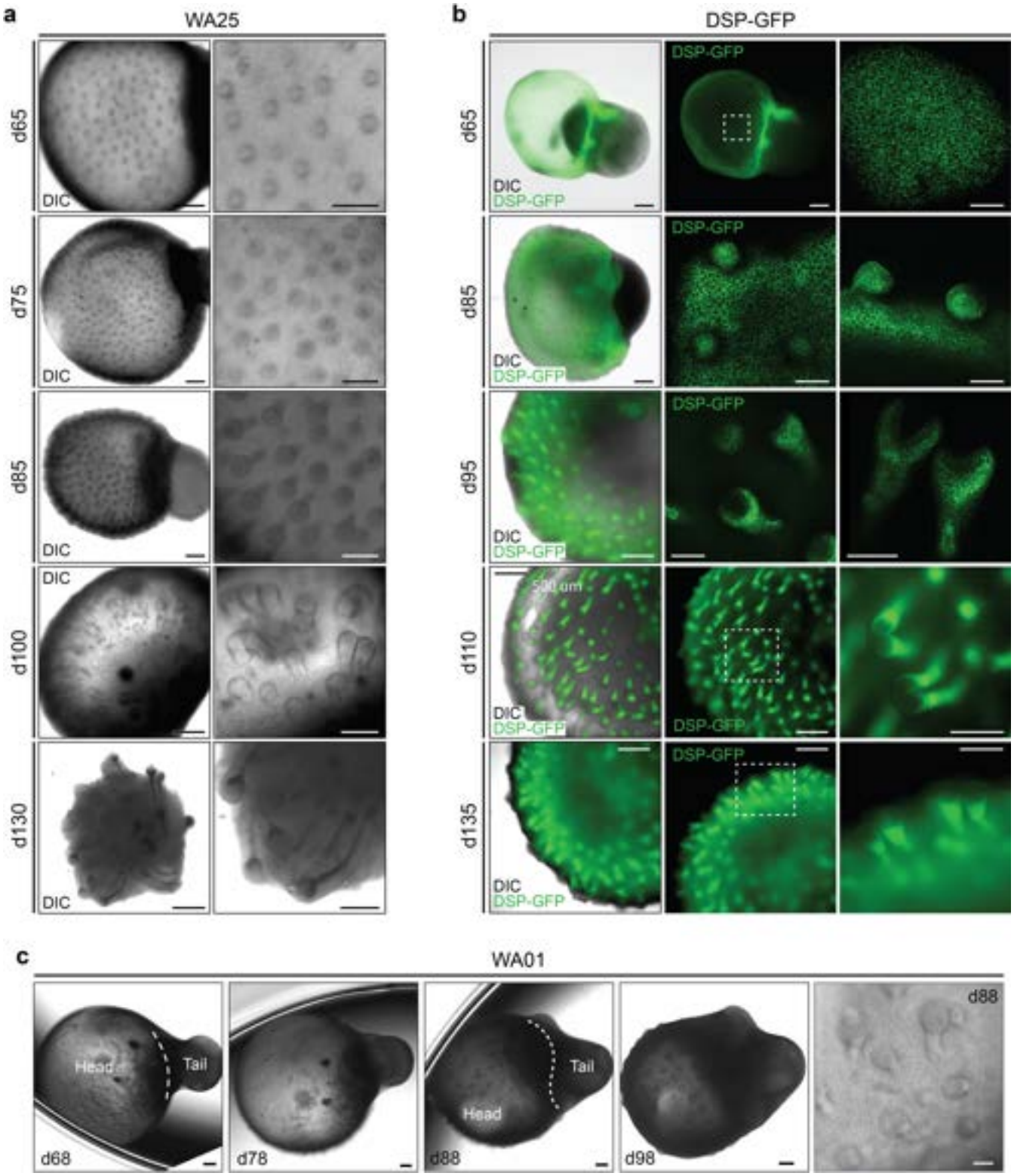


Extended Data Fig. 1 | See next page for caption.

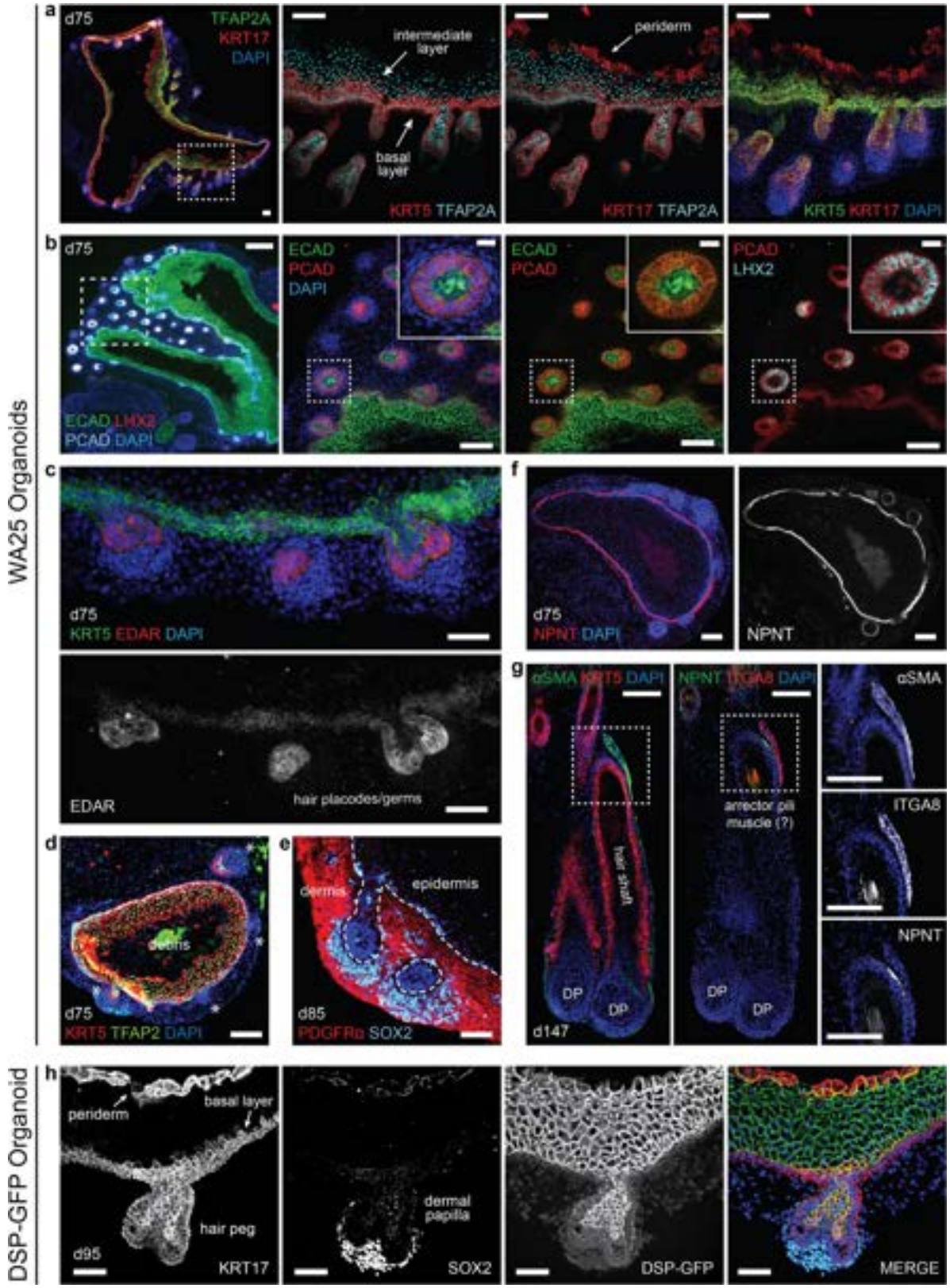
Extended Data Fig. 1 | Overview of skin organoid induction in relation to developmental events in normal human skin and morphological changes in skin organoids under different treatment regimes. a, Schematic overview of the skin organoid protocol. (i) Pre-aggregate human pluripotent stem cells to form an aggregate; (ii) induce differentiation of the aggregate to form surface ectoderm and CNC cells by modulating the TGFβ, BMP, and FGF signalling pathways; and (iii) provide an appropriate environment for the aggregate to mature into a complex skin organoid unit composed of fully stratified skin with a dermal layer that produces hair follicles, sensory neurons and cartilage. The cystic skin organoid can be transplanted and integrated into the back skin of a mouse as a planar layer, and can grow hair follicles. **b**, Comparison of the timeline of skin development in vitro and in vivo. Developmental timing is approximate. SkO, skin organoid. **c**, Representative differential interference contrast (DIC) and endogenous GFP fluorescence images of WA25 and DSP-GFP skin organoids on different days of differentiation under different treatment regimes. Top, WA25 skin organoids were differentiated in E6-based medium, treated with one of three different regimes on day 3—no LDN or FGF (d0BSF); LDN alone (d0BSF-d3L); and both LDN and FGF (d0BSF-d3LF)—and monitored during days 0–60. The aggregates that were treated with neither LDN nor FGF on day 3 maintained cystic organoids for about 30 days of differentiation, but the organoids lost their morphology and shrunk afterwards. Treatment with LDN alone on day 3 promoted cystic morphology, but was not sufficient to produce hair-bearing skin organoids in a consistent manner. Co-treatment with LDN and FGF on day 3 (the final optimized treatment condition) was optimal for epithelial stratification and sufficient development of the dermal layer. Tail-like regions containing mesenchymal and neuronal cells on one pole of the skin organoids are visible by day 18 of differentiation. Bottom, DSP-GFP skin organoids were also differentiated in E6-based medium and were treated on day 3 with both LDN and FGF (final optimized treatment regime). GFP^+ epithelium is visible on the surface or edge of the sphere-like organoids, and the GFP signal intensifies as the organoids differentiate and mature further. The tail portion of the organoid appears by day 22 of differentiation (the GFP^+ signal at the tail portion presented in the day-22 image is an autofluorescence). Right, WA25 skin organoids were differentiated in chemically defined medium (CDM)-based medium and treated with one of three different regimes on day 4—no LDN or FGF (NT); LDN alone (d4 LDN); and both LDN and FGF (d4LDN/FGF). By day 55 of differentiation, aggregates that were treated with neither LDN nor FGF on day 4 eventually lost their shape and shrunk. Treatment with LDN alone on day 4 maintained cystic organoid morphology, and occasionally induced maturation of skin organoids. Co-treatment with LDN and FGF on day 4 was suitable for skin organoids to fully mature in a relatively consistent manner. Changing the differentiation medium from CDM to E6 improved skin organoid development by reducing the tail portion (non-skin-related mesenchymal cells) and increasing the head (skin) portion. Optimization experiments were repeated at least three times independently. Scale bars, 500 µm. This figure corresponds with the data and concepts in Fig. 1.



Extended Data Fig. 2 | Induction of surface ectoderm and CNC cells in WA25 and DSP-GFP skin organoid cultures. a–**d**, Representative immunostaining images of day-16 WA25 (**a**, **b**) and DSP-GFP (**c**, **d**) aggregates with co-induced epithelium and CNC cells cultured in the optimized d0BSF-d3LF treatment conditions. P75 and SOX10 highlight neuroglia-associated CNC cells, and PDGFRα and TFAP2A highlight mesenchyme-associated CNC cells. In **a**, **b**, the epithelium is highlighted by ECAD and TFAP2A double-positive signals. In **c**, **d**, ECAD was omitted to allow observation of the endogenous DSP-GFP signal at the apical surface of the epithelium. Dashed boxes indicate magnified regions (right). Dashed lines outline the borders between the epithelium and CNC cells and between neuroglia- and mesenchyme-associated CNC cells. Note that there are non-specific fluorescence background noise signals from Matrigel. The immunostaining was repeated three times independently on a total of nine aggregates from three separate experiments. Scale bars: 100 μm (main images), 50 μm (magnified images). This figure corresponds with the data and concepts in Fig. 1c.

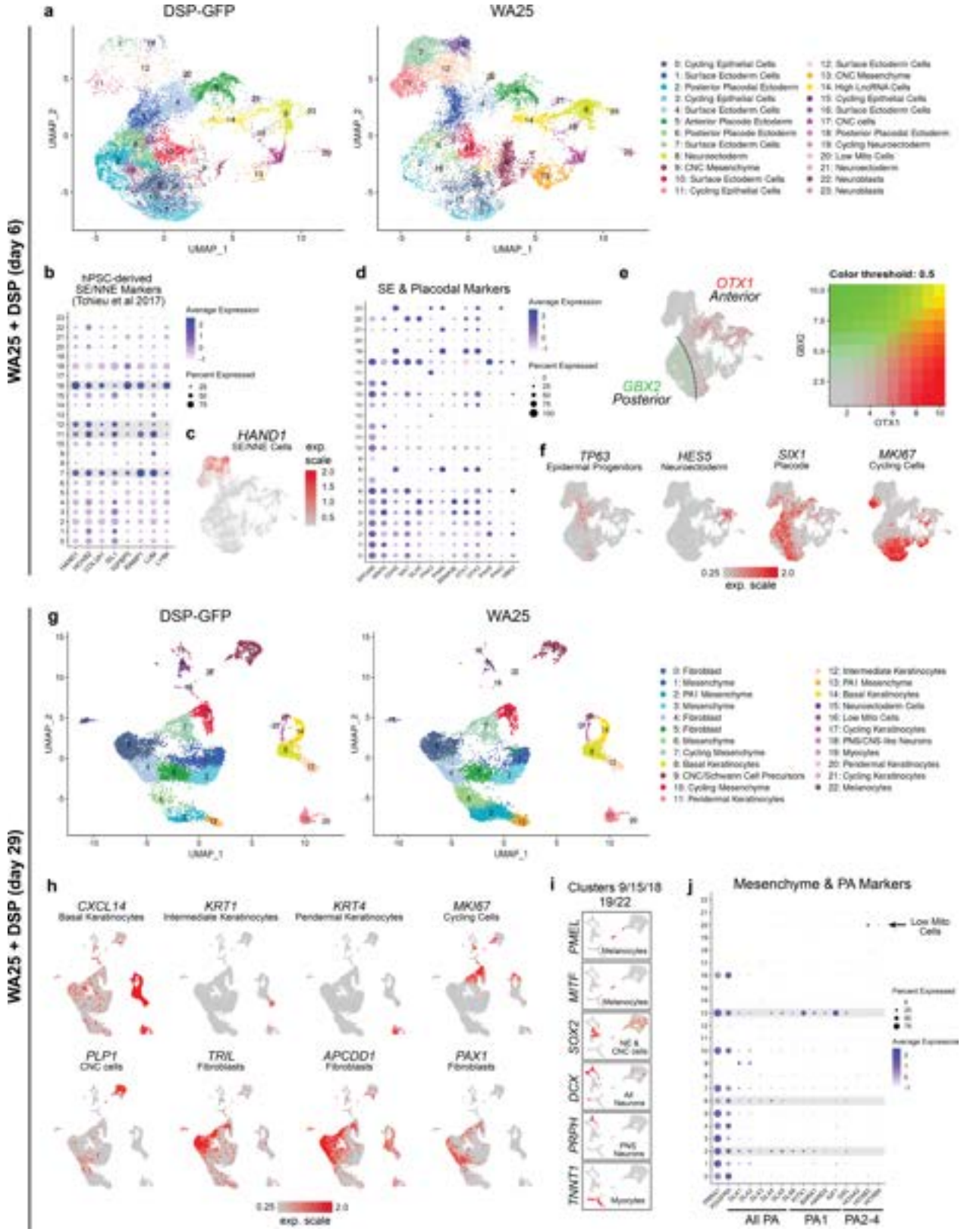


Extended Data Fig. 3 | Comparison of initial hair follicle induction in WA25 and DSP-GFP live-cell aggregates. a, Representative DIC images of WA25 skin organoids at days 65–130 with developing hair follicles. Scale bars, 500 µm (left); 250 µm (right). **b**, DIC and endogenous GFP fluorescence images of DSP-GFP skin organoids at days 65–135 with developing hair follicles. Dashed boxes indicate magnified regions (right). Scale bars, 500 µm (left, middle); 250 µm (far right, days 95, 110 and 135); 100 µm (far right, days 65 and 85). **c**, DIC images of WA01 skin organoids at days 68–98 with developing hair follicles. Right, magnified view of day-88 WA01 skin organoid hair follicles. The head (skin cyst) and tail (non-skin mesenchymal) portions within skin organoids are distinguishable. In addition, the hair bulbs are facing outward from the organoid, contacting the medium, whereas the hair shafts grow inward towards the centre of the skin organoid. Scale bars, 250 µm (all panels except far right); 50 µm (far right). The WA25 and DSP-GFP skin organoid images represent morphologies observed throughout nine independent cultures, and the WA01 skin organoid images represent one experiment performed by the Stanford group. This figure corresponds with the data in Fig. 1e, f.



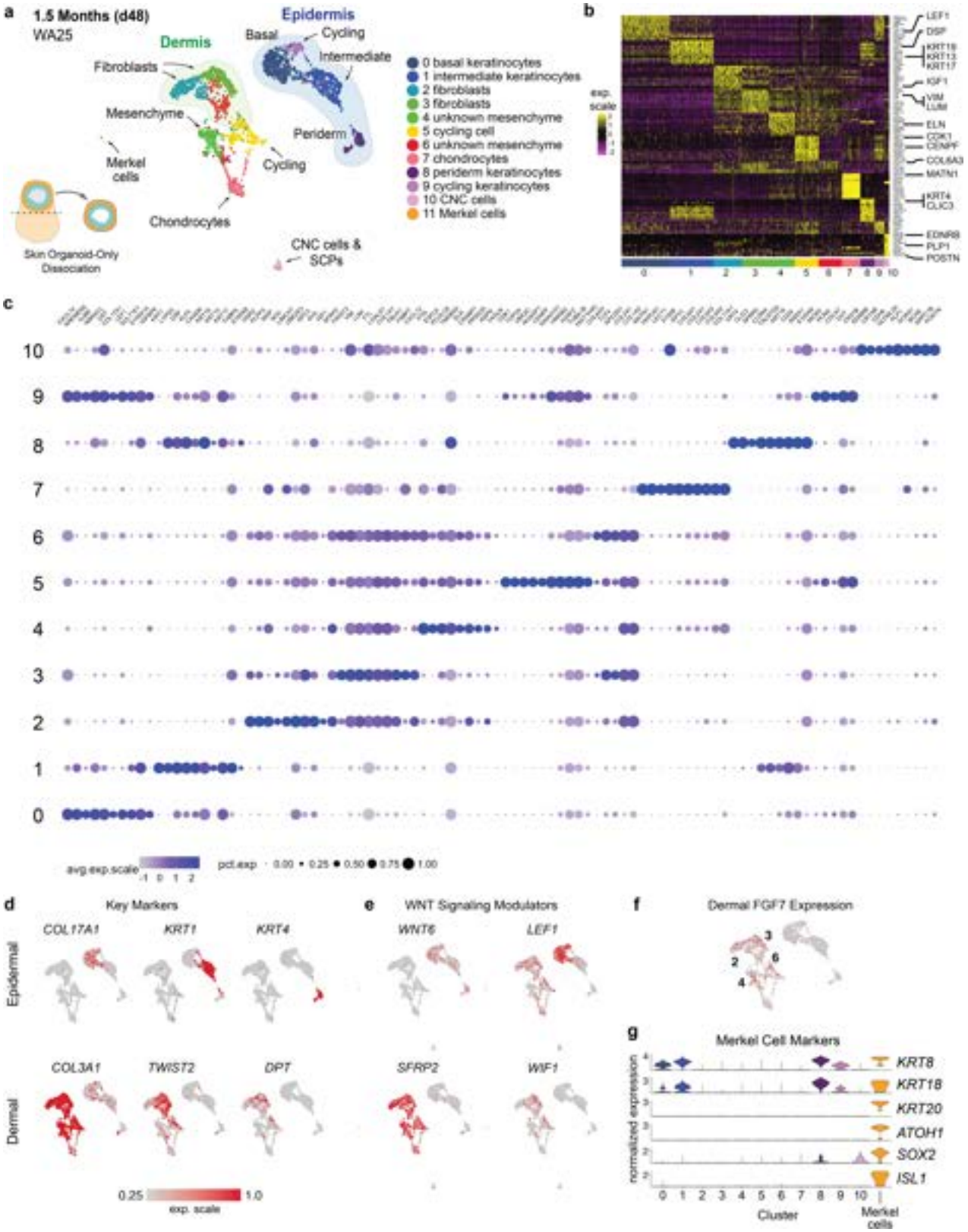
Extended Data Fig. 4 | See next page for caption.

Extended Data Fig. 4 | Key protein markers of hair follicle induction in skin organoids. a, A day-75 WA25 skin organoid with nascent hair follicles immunostained for KRT5, KRT17 and TFAP2A. This KRT17 antibody labels basal and peridermal keratinocytes. KRT5 and TFAP2A double-positive staining represents the basal layer, and TFAP2A single-positive staining between basal and peridermal layers indicates the intermediate layer. Dashed boxes indicate magnified regions. Scale bars, 100 µm. **b**, Immunostaining for ECAD in day-75 WA25 skin organoids labels the entire epithelium, whereas PCAD expression is restricted to the basal layer and the hair germ epithelium. LHX2 labels hair placodes and matrix cells. Dashed boxes indicate magnified regions of the hair germ images (top right insets). Scale bars, 250 µm (far left); 100 µm (right three panels); 23 µm (insets). **c**, Immunostaining for EDAR in day-75 WA25 skin organoids labels placodes and matrix cells of hair follicles. Scale bars, 50 µm. **d**, Low-magnification immunostaining image of a day-75 WA25 organoid shows the distribution of KRT5 and TFAP2A expression, representing basal and intermediate layers. The non-specific fluorescence signal in the core of the skin cyst is cell debris. Asterisks indicate developing hair follicle placodes and germs. Scale bar, 100 µm. **e**, Representative immunostaining image for PDGFRα and SOX2 in a day-85 WA25 skin organoid. PDGFRα is expressed throughout the dermis that has developed in the outer layer of the skin cyst, and SOX2 labels dermal condensate and dermal papilla cells, yet is also expressed in some basal epidermal cells—probably Merkel cells. Scale bar, 50 µm. **f**, By day 75 of differentiation, NPNT is localized to the basement membrane of the epithelium of WA25 skin organoids. Scale bars, 100 µm. **g**, Immunostaining image of a day-147 WA25 skin organoid. Expression of αSMA (alpha smooth muscle actin) and ITGA8 (integrin alpha 8)—characteristics of arrector pili muscle—was detected next to the $NPNT^+$ hair follicle bulge region, suggesting that our skin organoid may be capable of producing arrector pili muscle. However, this feature was observed extremely rarely in our present culture conditions, which are still at an early developmental stage, and further optimization of the medium will be necessary for long-term culture. Dashed boxes indicate magnified regions of the bulge with an arrector pili muscle-like phenotype (right). Scale bars, 100 µm. **h**, Immunostaining of a day-95 DSP-GFP skin organoid showing a hair peg with $KRT17^+$ peridermal layer and outer root sheath, $SOX2^+$ dermal papilla and GFP^+ desmosome-rich intermediate epidermis of the hair follicle. Scale bars, 50 µm. All immunostaining was repeated 3–5 times on 9–12 skin organoids generated from 4 independent experiments before selection of representative images. This figure corresponds with the data in Fig. 1.



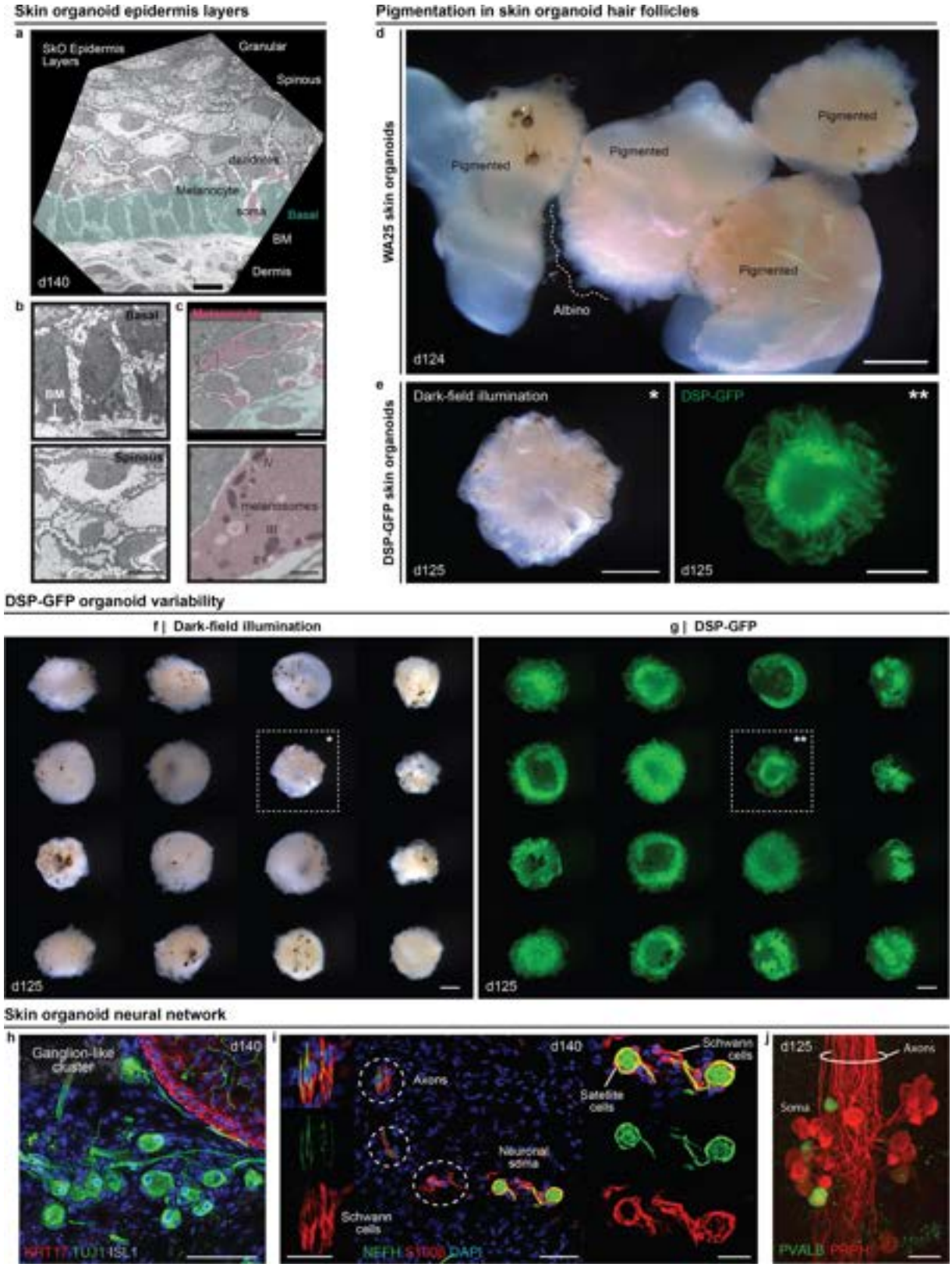
Extended Data Fig. 5 | See next page for caption.

Extended Data Fig. 5 | scRNA-seq analysis of day-6 and day-29 skin organoids derived from WA25 and DSP-GFP cells. a, **g**, Separate UMAP plots for WA25 and DSP-GFP cell clusters at day 6 (**a**) and day 29 (**g**). The major cell cluster groupings of surface epithelia, epidermis and mesenchyme are noted. Colours indicate cell state. The presumptive cell identities (based on a priori knowledge of marker genes) are listed. Cell clusters with no discernible identity had low expression of mitochondrial genes ('low mito cells') or high expression of long noncoding RNA genes ('high lncRNA'). Day 6 (**a**): $n = 11{,}785$ WA25 cells, $n = 11{,}544$ DSP-GFP cells; day 29 (**g**): $n = 9{,}268$ WA25 cells, $n = 9{,}013$ DSP-GFP cells. Ten day-6 organoids from one experiment and five day-29 organoids from one experiment were randomly pooled for scRNA-seq analysis (per cell line). **b**, Dot plot for surface ectoderm and non-neural ectoderm (NNE) markers on the basis of RNA-seq data from a previous study[11]. **c**, UMAP plot for *HAND1*, a key marker for NNE cells derived from human pluripotent stem cells. **d**, Dot plot for markers of surface ectoderm and anterior–posterior placodes. Gene expression frequency is indicated by spot size and expression level is indicated by colour intensity. **e**, UMAP overlay plot showing the distribution of *OTX1* (marker of anterior placode and neuroectoderm) and *GBX2* (marker of posterior placode). **f**, UMAP plots for key markers of epidermal progenitors, neuroectoderm, general placode and cycling cells. **h**, UMAP plots for specific marker genes that define epidermal, cycling, CNC and dermal cell subtypes. **i**, Magnified view of clusters 9 (CNC cell and Schwann cell precursors), 15 (neuroectoderm cells), 18 (PNS- or CNS-like neurons), 19 (myocytes) and 22 (melanocytes). Key marker genes are shown to label each cell cluster. *SOX2* has broad expression across neuroectoderm and CNC cells. **j**, Dot plot for general mesenchymal markers (*PRRX1* and *PDGFRA*), markers of all pharyngeal arches, markers of PA1 and markers of PA2–PA4. Gene expression frequency is indicated by spot size and expression level is indicated by colour intensity. The expression of HOX genes appears to be limited to a subset of cells with low mitochondrial gene expression. This figure corresponds with the data in Fig. 2.



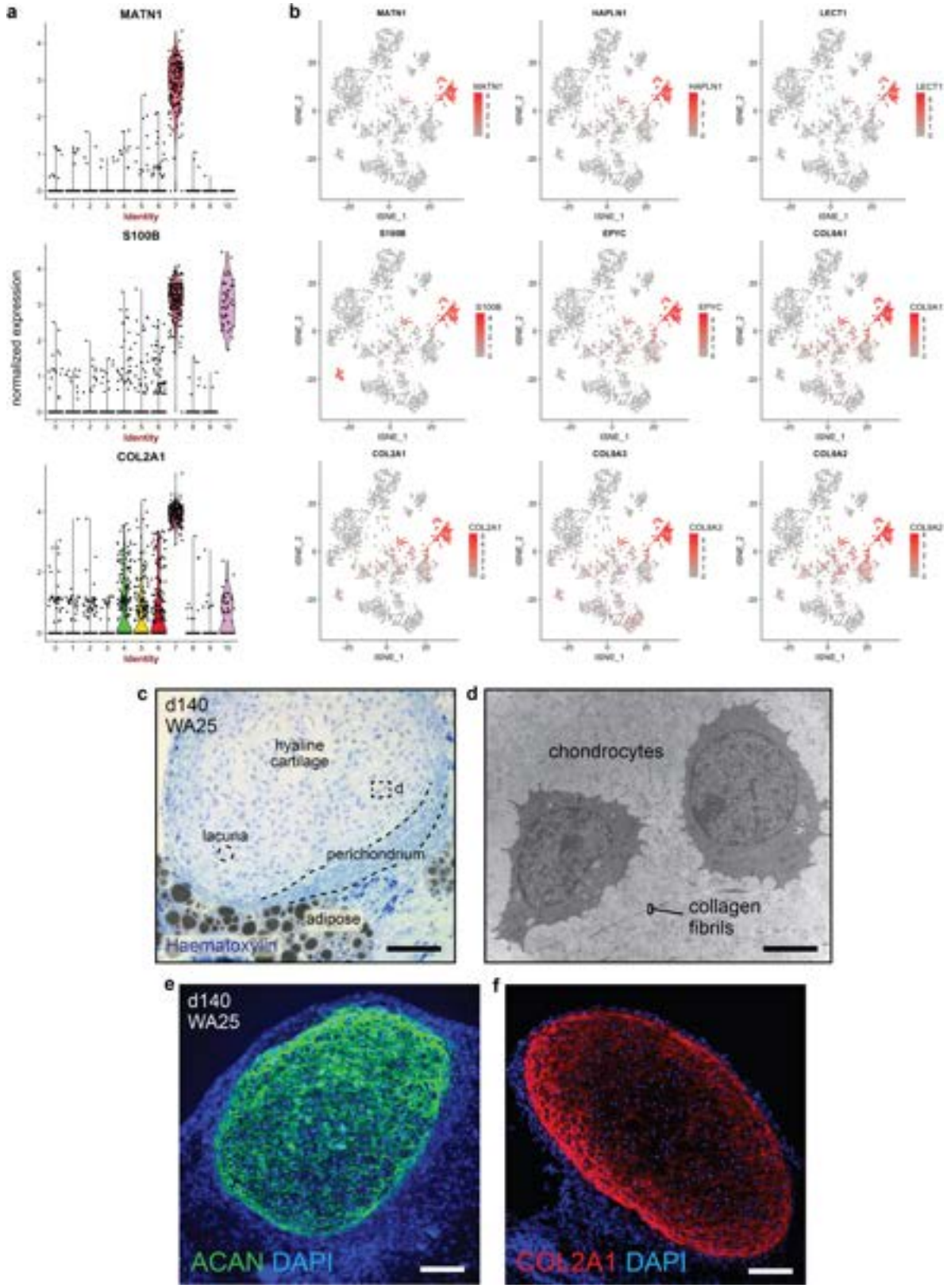
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Extended Data Fig. 6 | scRNA-seq analysis of day-48 skin organoids derived from WA25 cells. a, UMAP clustering of day-48 WA25 cell subtypes. Colours indicate cell state. *n* = 2,491 cells. Six day-48 skin organoids from one experiment were pooled for scRNA-seq analysis. SCPs, Schwann cell precursor cells. **b**, Heat map showing the scaled log-normalized expression of the top 10 differentially expressed genes per cell cluster for the day-48 WA25 scRNA-seq dataset. *ELN* is also known as *SMIM6*. **c**, Dot plot array showing the top 10 positively expressed genes per cell cluster for the day-48 WA25 scRNA-seq dataset. Gene expression frequency is indicated by spot size and expression level is indicated by colour intensity. **d**, UMAP plots for marker genes of specific cell subtypes. **e**, UMAP plots for WNT signalling pathway genes. *WNT6* is expressed in basal keratinocytes and peridermal keratinocytes. *LEF1* expression appears localized to basal keratinocytes. Negative WNT modulatory genes, *SFRP2* and *WIF1*, are expressed in putative dermal fibroblasts of the mesenchymal cell group. **f**, Dermal fibroblast clusters also contain cells that express *FGF7* (keratinocyte growth factor). The numbers of cell clusters with positive expression are listed on the UMAP plot. **g**, Identification of Merkel cells. Using the UMAP clustering algorithm, we identified a subset (*n* = 8 cells) of cells of cluster 0 (putative basal keratinocytes) that were completely separated from most cluster-0 cells, suggesting that our unbiased analysis pipeline failed to identify a unique subset of low-abundance cells. We used the Seurat manual selection tool to generate an 11th cluster containing these cells (**a**, left). Violin plots show normalized gene expression of the Merkel cell marker genes *ATOH1*, *ISL1*, *SOX2*, *KRT8*, *KRT18* and *KRT20*.

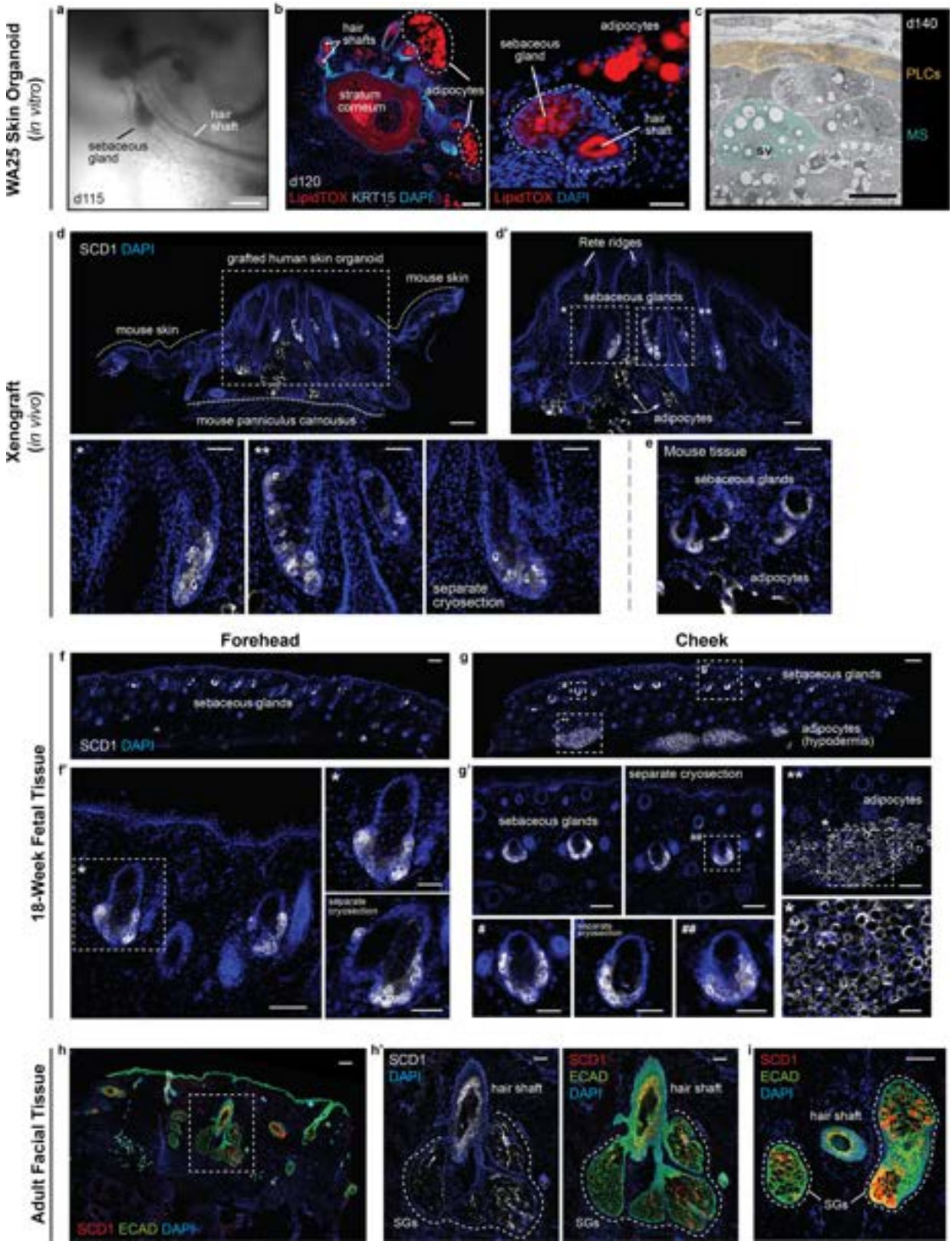


Extended Data Fig. 7 | See next page for caption.

Extended Data Fig. 7 | Pigmentation in hair follicles of WA25 and DSP-GFP skin organoids. a, TEM image of epidermis in a day-140 WA25 skin organoid. A melanocyte cell body is pseudo-coloured pink, and the basal skin layer is pseudo-coloured green. BM, basement membrane. Scale bar, 10 µm. **b**, Higher-magnification image of basal keratinocytes and the basement membrane (top) and spinous layer keratinocytes (bottom). Scale bars, 5 µm. **c**, Higher-magnification image of matrix-associated melanocytes (pink) with melanosomes at different stages (I, II, III and IV). The region in the dashed box is magnified in the lower panel. Scale bars, 5 µm (top); 800 nm (bottom). The images in **c** correspond to Fig. 3f. TEM was performed on two day-140 skin organoids from separate experiments (**a**–**c**). **d**, Dark-field illumination image of day-124 WA25 skin organoids, comparing the development of pigmented and non-pigmented (albino) hairs. Images represent pigmented and non-pigmented hairy skin organoids from nine independent experiments. Scale bar, 1 mm. **e**, Representative dark-field illumination (left, one asterisk) and endogenous GFP fluorescence (right, two asterisks) images of day-125 DSP-GFP skin organoids with pigmented hair follicles (dashed boxes in **f**, **g**). Scale bars, 1 mm. **f**, **g**, Overview of representative 16 out of 24 skin organoids from one DSP-GFP experiment. Each organoid exhibits pigmented hair follicles; however, the morphology of the epidermal cyst, as shown by DSP-GFP expression (green), varies between organoids. Magnified views of the organoids highlighted with dashed boxes and asterisks are shown in **e**. Scale bars, 1 mm. Images show a set of organoids from one experiment and are representative of morphologies observed over the course of nine independent experiments (**e**–**g**). **h**, Immunostaining for TUJ1 and ISL1 in a day-140 WA25 organoid reveals the ganglion-like cluster of neurons. Scale bar, 50 µm. **i**, A subset of organoid neurons express NEFH and are associated with S100β$^+$ satellite glial cells and Schwann cells. Dashed circles indicate axons of neurons. Scale bars, 50 µm (main image); 25 µm (inset). **j**, Immunostaining for PVALB in a day-125 WA25 organoid reveals the proprioceptors. PRPH$^+$ sensory neurons have small somas, and their axons form a fascicle. Scale bar, 35 µm. Immunostaining was repeated at least three times on five or six independent organoids (**h**–**j**). This figure corresponds with the data in Fig. 3.

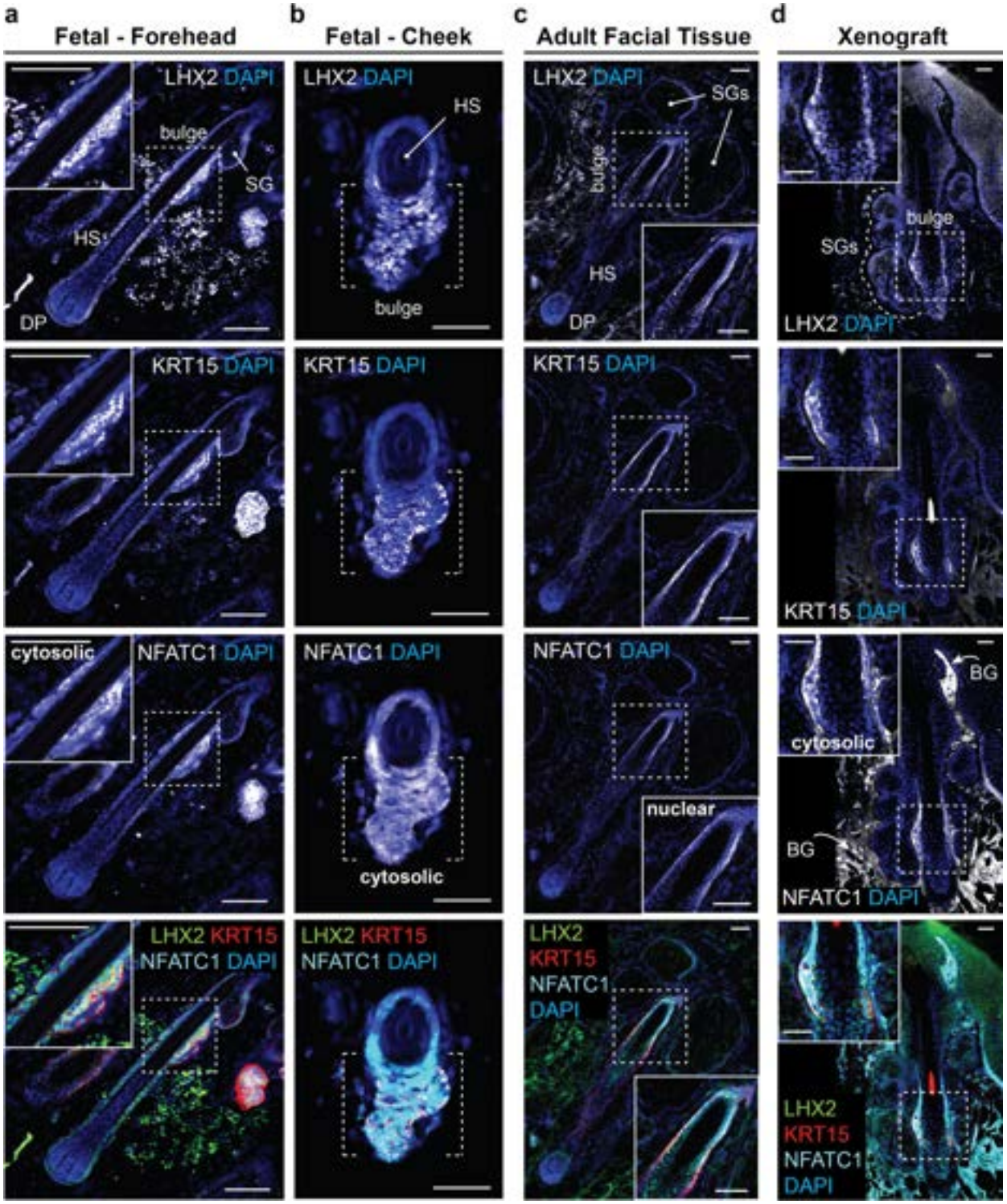


Extended Data Fig. 8 | Chondral development in the tail region of skin organoids. a, **b**, Violin and *t*-SNE plots showing normalized expression of chondral marker genes within cluster 7 of the day-48 WA25 skin organoid dataset (Extended Data Fig. 6, Supplementary Data 7). *LECT1* is also known as *CNMD*. The data represent cells pooled from six day-48 skin organoids from one experiment. **c**, Haematoxylin-stained section of a day-140 skin organoid showing hyaline cartilage that has formed within skin organoid-associated mesenchymal tissue. Three independent haematoxylin stainings were performed on six skin organoids from three different experiments. **d**, TEM image of two representative chondrocytes located within hyaline cartilage tissue (dashed box in **c**). TEM was performed once on two different skin organoids. **e**, **f**, Immunostaining of day-140 organoid samples for aggrecan (ACAN) and collagen type II alpha 1 (COL2A1) highlights cartilage development. Images represent one of three independent IHC stainings on six skin organoids produced from three separate experiments. Scale bars, 100 µm (**c**–**f**). This figure corresponds with the data in Figs. 2, 3.



Extended Data Fig. 9 | See next page for caption.

Extended Data Fig. 9 | Hair follicles of xenografted WA25 skin organoids have sebaceous glands comparable to second-trimester (18 weeks) fetal and adult facial hair. a, Bright-field image of two day-115 (in vitro) WA25 skin organoid hair follicles with visible hair shafts and sebaceous glands. Image represents hair follicles produced in nine independent experiments. Scale bar, 250 µm. **b**, Left, LipidTOX staining of day-120 WA25 skin organoid hair follicles reveals lipid-rich cells such as sebocytes and adipocytes (outlined by dashed lines). Immunostaining for KRT15 labels the outer root sheath of hair follicle cells. Right, high-magnification image of an adjacent cryosection of the specimen in the left panel. Dashed line highlights a horizontally cross-sectioned hair shaft and a sebaceous gland. Scale bars, 100 µm (left); 50 µm (right). **c**, TEM image of a sebaceous gland in a day-140 WA25 skin organoid. TEM was performed once on two different skin organoids from separate experiments. Peripheral layer cells (PLCs) and a maturing sebocyte (MS) containing sebum vacuoles (SV) have been pseudo-coloured in yellow and green, respectively. Scale bar, 5 µm. **d**, $SCD1^+$ sebaceous glands in xenografted day-140 WA25 skin organoids. Xenografts were extracted and fixed in 4% PFA, more than 49 days after transplantation. Dashed boxes and asterisks indicate the magnified regions in the subsequent images. Dashed lines distinguish between the tissue of the grafted human skin organoid and the tissues of the host NU/J mouse. Arrows indicate $SCD1^+$ adipocytes. Scale bars, 100 µm (**d**, **d'**); 50 µm (*, **, separate cryosection). **e**, $SCD1^+$ sebaceous glands in the NU/J mouse skin adjacent to xenografts. The size of mouse sebaceous glands is smaller than that of human sebaceous glands; the origin of sebaceous glands is distinguishable within the extracted xenograft samples. Scale bar, 50 µm. **f**, $SCD1^+$ sebaceous glands in 18-week human fetal forehead skin. Scale bars, 250 µm (**f**); 100 µm (**f'**); 50 µm (*, separate cryosection). **g**, $SCD1^+$ sebaceous glands and adipocytes in 18-week human fetal cheek skin. The adipocytes are prominently abundant in fetal cheek tissue compared to forehead tissue (**f** versus **g**). Dashed boxes and symbols highlight the magnified regions in the subsequent images. Scale bars, 250 µm (**g**); 100 µm (**g'**, **); 50 µm (*, #, ##, separate cryosection). **h**, **i**, $SCD1^+$ sebaceous glands in adult human facial skin. The epithelium is visualized by ECAD immunostaining. Dashed lines outline several lobes of the sebaceous glands of a hair follicle. Scale bars, 250 µm (**h**); 100 µm (**h'**, **i**). Immunostaining images represent one of five independent stainings on three to five different samples per tissue type. This figure corresponds with the data in Fig. 4.



Extended Data Fig. 10 | Hair follicles of xenografted WA25 skin organoids have bulge regions comparable to second-trimester (18 weeks) fetal and adult facial hair. a–**d**, Representative immunostaining images for markers of hair follicle stem cells (LHX2, KRT15 and NFATC1) in the hair follicle bulge region in 18-week human fetal skin from two facial locations (forehead and cheek) (**a**, **b**), adult facial skin (**c**) and xenografted skin organoid tissue (**d**). In both fetal and xenograft hair follicles, NFATC1 expression is predominantly localized to the cytoplasm in bulge cells, whereas in adult facial hair follicles, NFATC1 expression is nuclear-localized in the hair follicle bulge region—consistent with previous reports of nuclear-localized NFATC1 in mouse bulge stem cells[36]. Arrows indicate background (BG) staining noises. Dashed boxes in **a**, **c**, **d** indicate magnified bulge regions, which are presented in a corner of each image. Dashed brackets in **b** indicate the bulge region. Representative immunostaining images are selected from five independent stainings on three to five different samples per tissue type. Scale bars, 100 μm (**a**, **c**); 50 μm (**b**, **d**). This figure corresponds with the data in Fig. 4.

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used to collect data.
Data analysis	For single-cell analysis we used 10x Genomics CellRanger and Seurat v2 and v3 and RStudio. For statistical analysis we used GraphPad Prism 8. For image analysis we used ImageJ (FIJI), Leica LAS X Software (v3), and Imaris 8.

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Single-cell RNA sequencing data have been uploaded to the Gene Expression Omnibus (Accession code GSE147206): https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE147206. In the supplementary material, we have provided HTML files that generate an online interface with which to explore our scRNA-seq analysis pipeline and evaluate additional cell cluster markers for the integrated and non-integrated datasets (Supplementary Data 1-7). In addition, the integrated datasets shown in Figure 2 can be queried via a data exploration portal hosted on our laboratory website: https://www.koehler-lab.org/resources. Source data behind Figures 1-4 are available within the manuscript files.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No statistical methods were used to determine sample size. Generally, we sought to technically replicate experiments >6 times over >3 separate experiments. These numbers were chosen based on previous studies in the organoid literature.
Data exclusions	No data were excluded.
Replication	The number of replications is noted throughout the manuscript. Also, see specific sections on reproducibility in the METHODS. For histology, we analyzed multiple organoids from each cell line at different timepoints to demonstrate the robustness of the technique. All replication attempts were successful unless noted, as in our Supplementary Note on methodology development.
Randomization	In general, the experiments were not randomized; however, organoid specimens used for histology and single-cell RNA-sequencing analysis were selected at random and not screened for quality or specific morphological characteristics.
Blinding	The investigators were not blinded during experiments and outcome assessments. The same investigators performed experiments and data analyses, which made blinding difficult. Our use of single-cell RNA-sequencing and unbiased clustering analysis helped to mitigate any potential bias from a lack of blinding.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☐	☒ Animals and other organisms
☐	☒ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Antibody Name (Species, Company, Cat. No., Dilution)
Aggrecan (Rabbit, Invitrogen, PA1-1745, 1:100)
alpha-smooth muscle actin (Mouse, Abcam, ab7871, 1:50)
βIII-Tubulin (TUJ1) (Mouse BioLegend 801202 1:100)
CD49f (Rat Invitrogen 12-0495-81 1:50)
Collagen 2A1 (COL2A1) (Mouse, Invitrogen, MA1-37493, 1:10)
Cytokeratin 5 (KRT5) (Rabbit, Invitrogen, MA5-14473, 1:50)
Cytokeratin 5 (KRT5) (Mouse, Invitrogen, MA5-12596, 1:50)
Cytokeratin 10 (KRT10) (Mouse, Santa Cruz, sc-23877, 1:50)
Cytokeratin 15 (KRT15) (Mouse, Santa Cruz, sc-47697, 1:50)
Cytokeratin 17 (KRT17) (Mouse, Santa Cruz, sc-393091, 1:50)
Cytokeratin 20 (KRT 20) (Rabbit, Cell Signaling Technology, 13063S, 1:100)
Cytokeratin 71 (KRT 71) (Rabbit, Invitrogen, PA5-83965, 1:500)
Cytokeratin 74 (KRT 74) (Rabbit, Sigma-Aldrich, HPA048596, 1:1000)
Cytokeratin 75 (KRT 75) (Rabbit, Invitrogen, PA553987, 1:50)
Cytokeratin 82 (KRT 82) (Rabbit, Sigma-Aldrich, HPA067230, 1:100)
E-cadherin (ECAD) (Mouse, BD Biosciences, 610181, 1:50)
EDAR (Goat, R&D Systems, AF745-SP, 1:50)
GATA3 (Mouse, Invitrogen, PIMA1028, 1:100)
Human nuclear antigen (Mouse, Abcam, ab191181, 1:100)

Integrin-alpha-8 (ITGA8) (Mouse, Santa Cruz, sc-365798, 1:50)
Islet1 (ISL1) (Mouse, DSHB, 39.4D5, 1:5)
Ki67 (Mouse, BD Biosciences, 550609, 1:100)
LHX2 (Rabbit, Millipore, ABE1402, 1:750)
Loricrin (Rabbit, Abcam, ab85679, 1:50)
Nephronectin (NPNT) (Mouse, R&D Systems, MAB8025, 1:50)
Neurofilament-heavy chain (NEFH) (Mouse, Cell Signaling Technology, 2836S, 1:100)
NFATC1 (Mouse, Invitrogen, MA3-024, 1:50)
P-cadherin (PCAD) (Mouse, Invitrogen, 32-4000, 1:50)
p75NTR (P75) (Rabbit, Cell Signaling Technology, 8238, 1:50)
pan-Cytokeratin (AE13) (Mouse, Santa Cruz, sc-57012, 1:50)
Parvalbumin (Mouse, Sigma-Aldrich, P3088, 1:1000)
PDGFRα (D13C6) (Rabbit, Cell Signaling Technology, 5241, 1:50)
Peripherin (Rabbit, Abcam, ab4666, 1:2500)
PMEL (Mouse, Novus, NBP2-29407, 1:100)
S100β (Rabbit, Abcam, ab52642, 1:75)
SCD1 (Rabbit, Sigma-Aldrich, HPA012107, 1:50)
SOX2 (Mouse, BD Biosciences, 561469, 1:100)
SOX10 (Mouse, Invitrogen, 14-5925-80, 1:50)
TFAP2A (Mouse, DSHB, 3B5, 1:5)

Validation

Previously published or manufacturer-validated antibodies were used for this study. For each antibody, the company website contains validation data. All antibodies were tested, in house, on PFA-fixed and cryosectioned human/mouse primary skin tissue samples.

Eukaryotic cell lines

Policy information about cell lines

Cell line source(s)	hESC lines: WA25 and WA09 (H9) - obtained from WiCell; hiPSC lines: mND2-0 (obtain from WiCell) and DSP-GFP (also known as AICS-0017, obtain from Coriell Institute)
Authentication	The cell lines were obtained under a materials transfer aggreement with the WiCell or Coriell Institute. Each line was determined to have a normal karyotype and the ability to generate all three germ layers at the time of derivation.
Mycoplasma contamination	The cell lines are negative for mycoplasma.
Commonly misidentified lines (See ICLAC register)	No ICLAC lines were used.

Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

Laboratory animals	For xenograft experiments, 5-6 week-old female immunodeficient nude mice (NU/J) were purchased from The Jackson Laboratory.
Wild animals	No wild animals were used for this study.
Field-collected samples	No field-collected samples were used for this study.
Ethics oversight	All mouse experiments were approved by the Indiana University School of Medicine Animal Care and Use Committee (IACUC), protocol number 11077.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Human research participants

Policy information about studies involving human research participants

Population characteristics	Human foetal and adult skin tissue samples were collected without any identifying information including sex, race, or gender.
Recruitment	No human participants were used in this study. Foetal tissue was obtain from misscarried specimens with the patients consent. Foetal tissue was procured by the University of Washington Birth Defects Research Laboratory. Adult human skin samples were procured by the Human Skin Disease Resource Center at Brigham and Women's Hospital-Harvard Medical School.
Ethics oversight	This work falls under NIH Exemption 4, thus it was designated as "Non-Human Subjects Research" by the Indiana University School of Medicine IRB.

Note that full information on the approval of the study protocol must also be provided in the manuscript.